

UMC 0.11um RFCMOS AI Advanced Enhancement Low Leakage 1.2V SPICE Model Document

Version 0.2 phase1

2011/03/07

Table of Contents

1	INTRODUCTION.....	5
1.1	Revision history	5
1.2	General Information.....	6
1.3	Model Usage:	8
1.3.1	For Cadence Spectre	8
1.3.2	For Synopsis Hspice	8
1.3.3	For Mentor Graphics Eldo	8
1.4	Device Architecture and Layout	9
1.4.1	Twin Well NMOS.....	9
1.4.2	PMOS.....	11
1.4.3	Triple Well NMOS	13
1.5	Model Features	15
2	RF MODEL VERIFICATION	16
2.1	Twin Well NMOS Model	16
2.1.1	Corner Table	16
2.1.2	Summary Table.....	20
2.1.3	Simulation results for WF, LF and NF trends	22
2.1.4	DC & RF Fitting curves for individual devices.....	24
2.2	PMOS Model	35
2.2.1	Corner Table	35
2.2.2	Summary Table.....	39
2.2.3	Simulation results for WF, LF trends	41
2.2.4	DC & RF Fitting curves for individual devices.....	43
3	APPENDIX.....	54
3.1	HSPICE Warning Message.....	54
3.2	SPECTRE Warning Message	54
3.3	ELDO Warning Message.....	54

List of Figures

Figure 1-1 Example layout of 1.2V pwell RF NMOS	9
Figure 1-2 Extension equivalent circuit of TWIN-Well NMOS	10
Figure 1-3 Example layout of 1.2V pwell RF NMOS	11
Figure 1-4 Extension equivalent circuit of PMOS.....	12
Figure 1-5 Example layout of Triple-Well RF NMOS	13
Figure 1-6 Sub-circuit topology of 1.2V Triple-Well RF NMOS	14
Figure 2-1 f_T and I_{dsat} vs. channel length	22
Figure 2-2 f_T and I_{dsat} vs. channel width	23
Figure 2-3 I_d - V_g (25°C).....	24
Figure 2-4 G_m - V_g (25°C)	25
Figure 2-5 I_d - V_d (25°C).....	26
Figure 2-6 R_{out} - V_d (25°C)	27
Figure 2-7 $Mag(H_{21})$ -Freq (25°C).....	28
Figure 2-8 f_T - V_g (25°C)	29
Figure 2-9 f_T - I_d (25°C).....	30
Figure 2-10 S_{11} (25°C)	31
Figure 2-11 S_{12} (25°C).....	32
Figure 2-12 S_{21} (25°C)	33
Figure 2-13 S_{22} (25°C).....	34
Figure 2-14 f_T and I_{dsat} vs. channel length	41
Figure 2-15 f_T and I_{dsat} vs. channel width	42
Figure 2-16 I_d - V_g (25°C).....	43
Figure 2-17 G_m - V_g (25°C)	44
Figure 2-18 I_d - V_d (25°C).....	45
Figure 2-19 R_{out} - V_d (25°C)	46
Figure 2-20 $Mag(H_{21})$ -Freq (25°C).....	47
Figure 2-21 f_T - V_g (25°C)	48
Figure 2-22 f_T - I_d (25°C).....	49
Figure 2-23 S_{11} (25°C)	50
Figure 2-24 S_{12} (25°C).....	51
Figure 2-25 S_{21} (25°C)	52
Figure 2-26 S_{22} (25°C).....	53

List of Tables

Table 1-1 Revision history	5
Table 1-2 Wafer Information	7
Table 2-1 RF & Logic Corner Table at 25 °C	17
Table 2-2 RF & Logic Corner Table at 125 °C	18
Table 2-3 RF & Logic Corner Table at -50 °C	19
Table 2-4 Summary Table for Twin Well RF NMOS	21
Table 2-5 RF & Logic Corner Table at 25 °C	36
Table 2-6 RF & Logic Corner Table at 125 °C	37
Table 2-7 RF & Logic Corner Table at -55 °C	38
Table 2-8 Summary Table for RF PMOS	40
Table 3-1 HSPICE warning message.....	54
Table 3-2 SPECTRE warning message	54
Table 3-3 ELDO warning message	54

1 Introduction

1.1 Revision history

Table 1-1 Revision history

Version	Date	Author	Remark
0.2_p1	2011/03/07	Daniel Wan Peiming Lei	1. Original release 2. Silicon-based model

1.2 General Information

Following Logic/MM model is used as the main body of the subckt RF MOSFET model and it would be fine tuned in order to improve Gm and Rout characteristics.

l110ae_ll12_v021_rf

More information about the model is specified below.

Model File Name:

Spectre: l110ae_ll12_v021_rf.lib.scs, l110ae_ll12_v021_rf.mdl.scs

Hspice: l110ae_ll12_v021_rf.lib, l110ae_ll12_v021_rf.mdl

Eldo: l110ae_ll12_v021_rf.lib.eldo, l110ae_ll12_v021_rf.mdl.eldo

Inline Sub-circuit Model Name & Node Definition:

Twin Well NMOS: n_12_llrf (drain gate source body)

PMOS: p_12_llrf (drain gate source body psub)

Triple Well NMOS: n_12bpw_llrf (drain gate source body nwell psub)

Corner Section:

tt : Typical n-ch MOSFET and Typical p-ch MOSFET

ff : Fast n-ch MOSFET and Fast p-ch MOSFET

ss : Slow n-ch MOSFET and Slow p-ch MOSFET

fnsp : Fast n-ch MOSFET and Slow p-ch MOSFET

snfp : Slow n-ch MOSFET and Fast p-ch MOSFET

Input Parameters:

lf : gate finger length in m

wf : gate finger width in m

nf : gate finger number

Geometry scaling:

NMOS:

Finger Length (lf): 0.12um ~ 0.36um

Finger Width (wf): 0.9um ~ 7.2um

Finger No (nf): 2~ 22

Support even and odd finger number

PMOS:

Finger Length (lf): 0.12um ~ 0.36um

Finger Width (wf): 1.6um ~ 9.6um

Finger No (nf): 2 ~ 22

Support even and odd finger number

Triple-well NMOS:

Finger Length (lf): 0.12um ~ 0.36um

Finger Width (wf): 0.9um ~ 7.2um

Finger No (nf): 2~ 22

Support even and odd finger number

Biasing coverage:

Voltage: Upto Vdd=1.2V

Frequency: upto 20.1GHz

Temperature coverage:

-50 C ~ +125 C

Simulator Versions:

Synopsys HSPICE release 2007.09

Cadence SPECTRE version 6.2.1.299

Mentor Graphics ELDO version 6.11_1.1

Wafer Information:

Table 1-2 Wafer Information

Mask ID	BSW0FA
Lot ID	D54AH
Wafer Number	23

1.3 Model Usage:

1.3.1 For Cadence Spectre

- Copy ModelFileName.lib.scs and ModelFileName.mdl.scs into your simulation directory.
- Add the following statement in the input netlist to the simulator
include "./ModelFileName.lib.scs " section = CornerSection, where CornerSection could be tt, ss, ff, snfp, fnsp.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
- SetOption1 options scale=0.9
- Follow the example below to call the model
x1 (drain gate source body) n_12_llrf lf=0.12e-6 wf=3.6e-6 nf=4 m=8

1.3.2 For Synopsis Hspice

- Copy ModelFileName.lib and ModelFileName.mdl into your simulation directory.
- Add the following statement in the input netlist to the simulator
.lib "./ModelFileName.lib " CornerSection, where CornerSection could be tt, ss, ff, snfp, fnsp.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
- Follow the example below to call the model
x1 drain gate source body n_12_llrf lf=0.12e-6 wf=3.6e-6 nf=4 m=8

1.3.3 For Mentor Graphics Eldo

- Copy ModelFileName.lib.eldo and ModelFileName.mdl.eldo & into your simulation directory.
- Add the following statement in the input netlist to the simulator
.lib "./ModelFileName.lib " CornerSection, where CornerSection could be tt, ss, ff, snfp, fnsp.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
- Follow the example below to call the model
x1 drain gate source body n_12_llrf lf=0.12e-6 wf=3.6e-6 nf=4 m=8

ModelFileName: l110ae_ll12_v021_rf

1.4 Device Architecture and Layout

1.4.1 Twin Well NMOS

- Double-side gate contacts are used to reduce gate resistance.
- Dummy Poly gates are used at each side to improve the Poly CD uniformity of poly fingers.
- The pick-ups of source and drain use M1-M3 metal stacks to improve current handling capabilities.
- The space between Poly gate and S/D contacts is 0.12um.

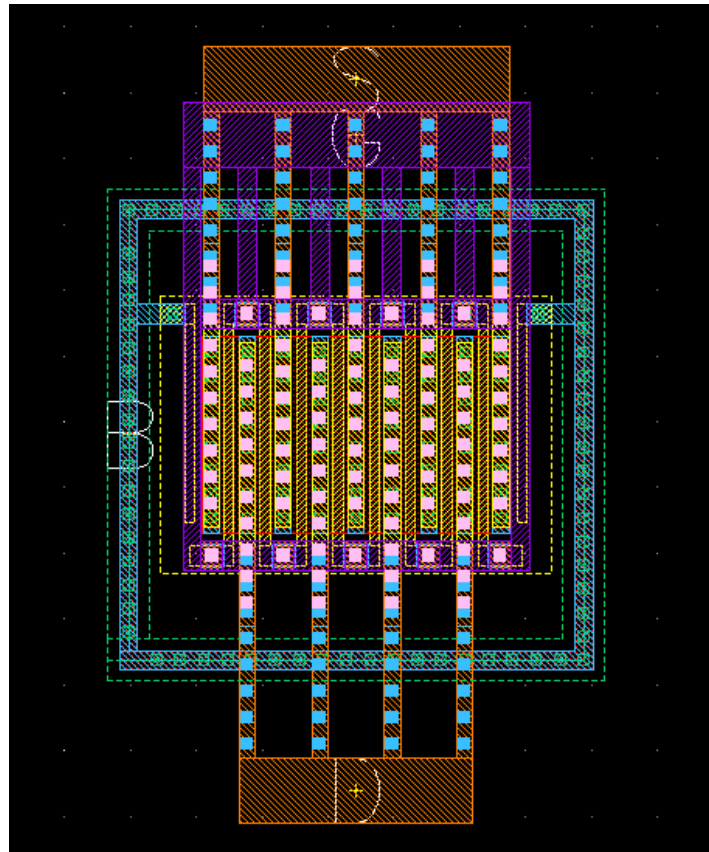


Figure 1-1 Example layout of 1.2V pwell RF NMOS

Extension equivalent circuit model:

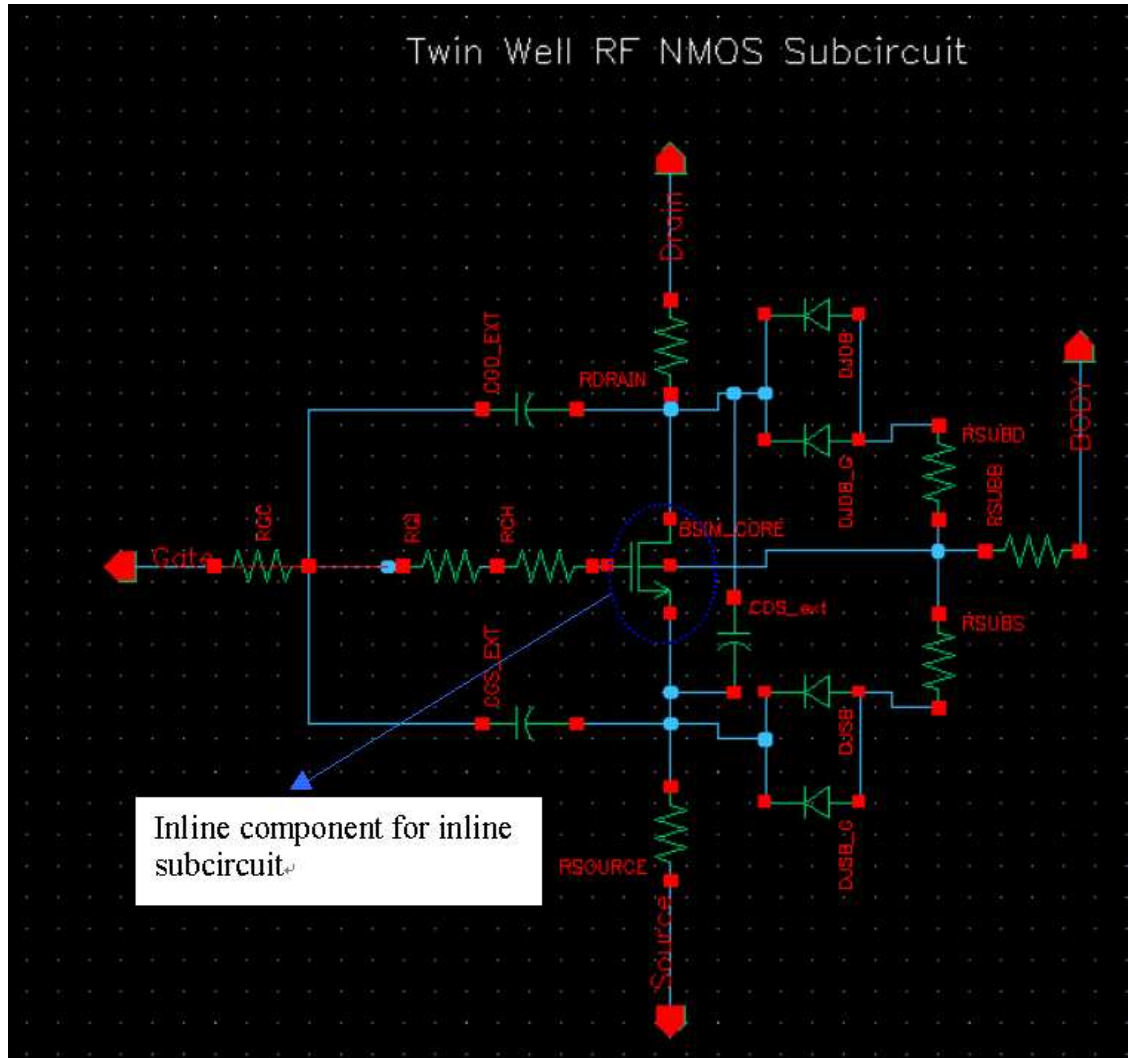


Figure 1-2 Extension equivalent circuit of TWIN-Well NMOS

1.4.2 PMOS

- Double-side gate contacts are used to reduce gate resistance.
- Dummy Poly gates are used at each side to improve the Poly CD uniformity of poly fingers.
- The pick-ups of source and drain use M1-M3 metal stacks to improve current handling capabilities.
- The space between Poly gate and S/D contacts is 0.12um.

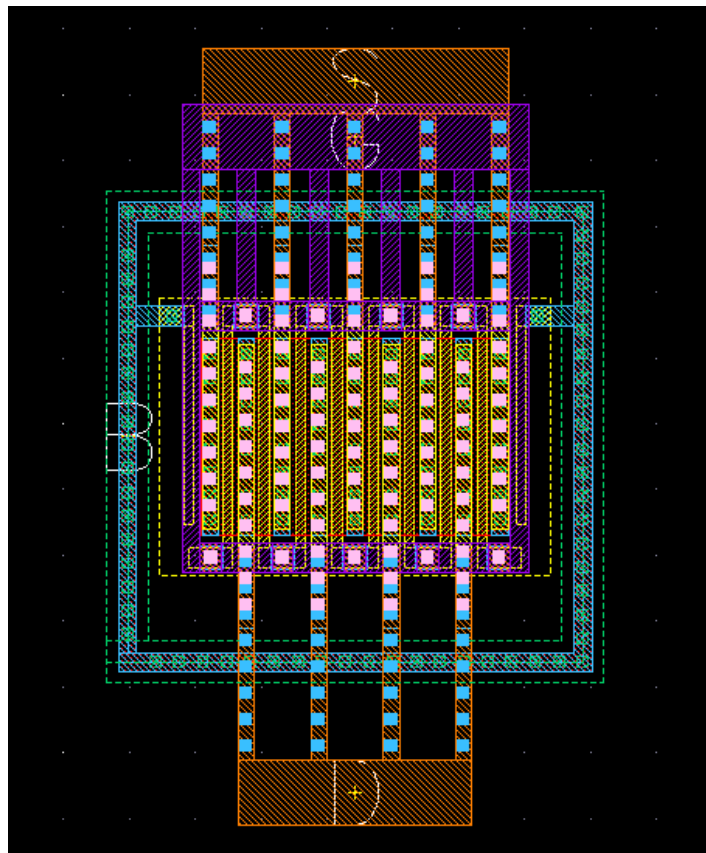


Figure 1-3 Example layout of 1.2V pwell RF NMOS

Extension equivalent circuit model:

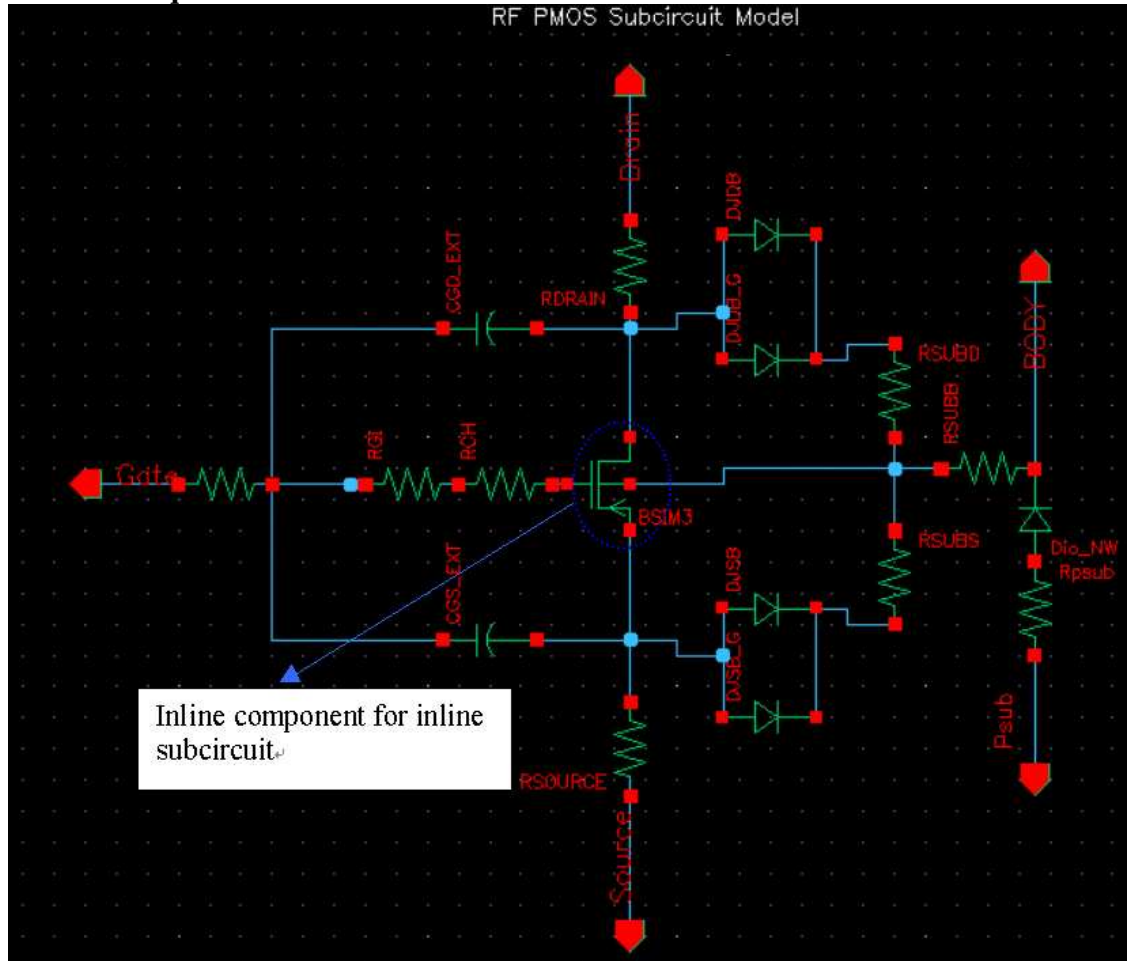


Figure 1-4 Extension equivalent circuit of PMOS

1.4.3 Triple Well NMOS

- Double-side gate contacts are used to reduce gate resistance.
- Dummy Poly gates are used at each side to improve the Poly CD uniformity of poly fingers.
- The pick-ups of source and drain use M1-M3 metal stacks to improve current handling capabilities.
- The space between Poly gate and S/D contacts is 0.12um.

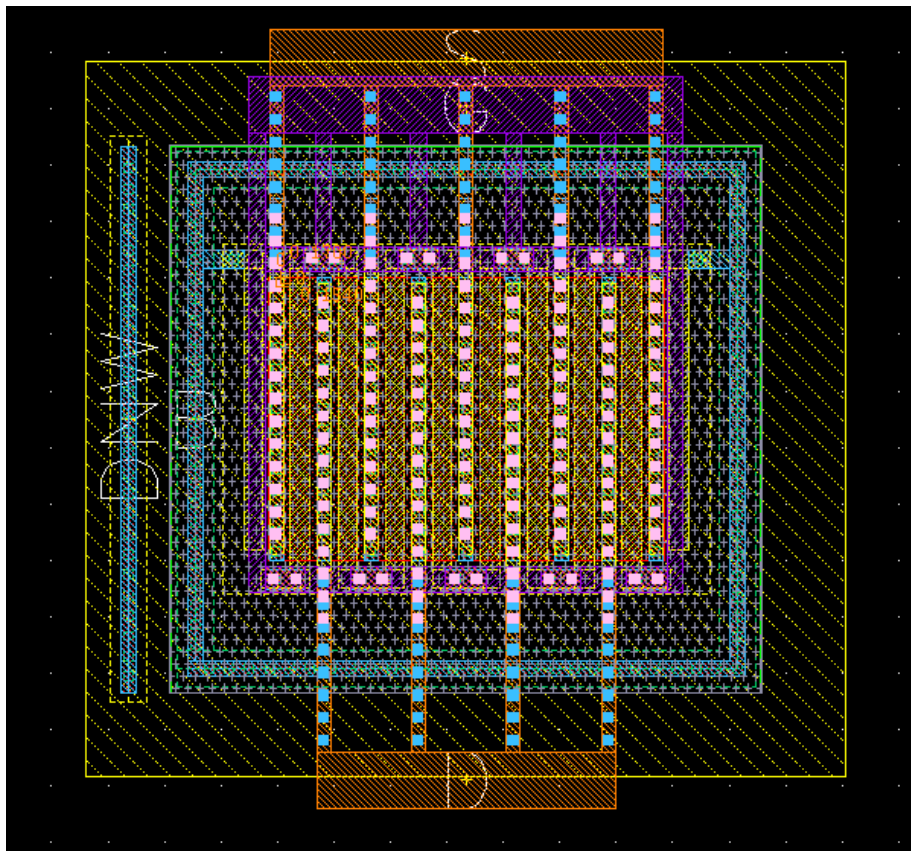


Figure 1-5 Example layout of Triple-Well RF NMOS

Extension equivalent circuit model:

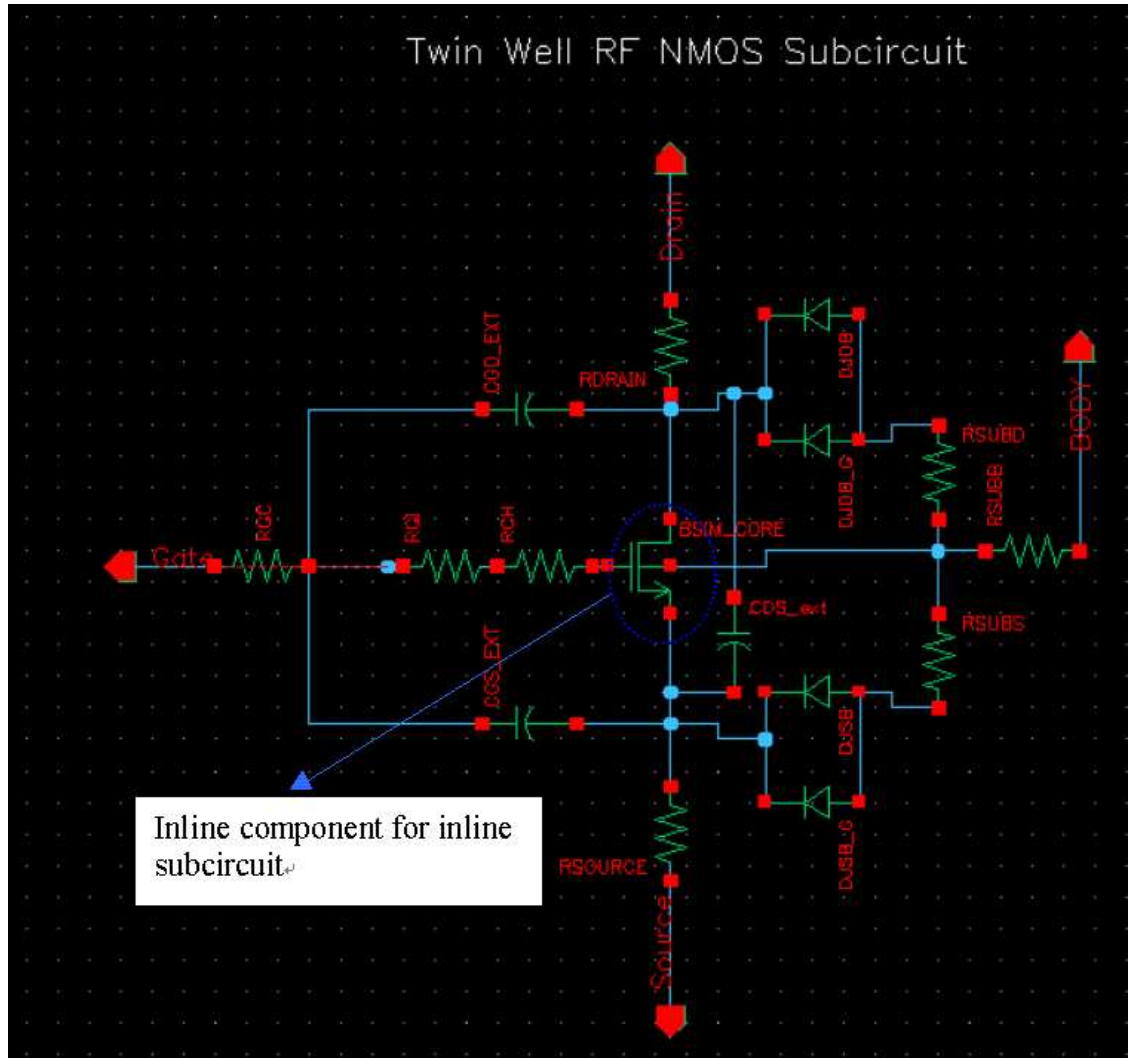


Figure 1-6 Sub-circuit topology of 1.2V Triple-Well RF NMOS

1.5 Model Features

BSIM3 :

The BSIM3model in RF subcircuit model is based on Logic/MM model, with parameters refined to improve the fitting quality of Gm/ Rout and RF performance.

LOD effect :

LOD model has been added to model STI stress & Finger number effects.

1/f Noise :

The BSIM3 1/f noise model is included

Note:

Model can support odd finger numbers, but it is recommended to use even finger number to reduce drain junction capacitance.

It is recommended not to use the model outside the above WF, LF and NF range. Outside above range, the model fitting quality may become worse.

This model should be used in conjunction with RF FET Pcell (Programmable layout Cell) provided in UMC MM/RF FDK (Foundry Design Kit). The model may not be valid if the transistor layout is different from UMC Pcell as shown in above layout example.

2 RF Model Verification

2.1 Twin Well NMOS Model

2.1.1 Corner Table

The tables below summarize the DC & RF simulation results for different corner cases for some selected devices and compared with Logic/MM model.

The biasing conditions for the parameters are:

ft : Cut-off frequency extracted @ Ft_max : Vd=1.2V, sweep Vg
vth : Linear Vth extracted @ Vd=0.05V @
Id=300nA*WF*0.9*NF/(LF*0.9+0.012um)
vtsat : Saturation Vth extracted @ Vd=1.2V using fixed current
Id=300nA*WF*0.9*NF/(LF*0.9+0.012um)
body effect : vtsat shift for Vbody =0 ~ -1.2V
Idsat : Drain current extracted @ Vd=Vg=1.2V
Idlin : Drain current extracted @ Vd=0.05V, Vg=1.2V
Ioff : Drain current extracted @ Vg=0V, Vd=1.2V

Temp=25C		RF Model				Logic Model			
Device	LF (um)	0.12	0.12	0.12	0.18	0.12	0.12	0.12	0.18
	WF (um)	7.2	3.6	0.9	2.4	7.2	3.6	0.9	2.4
	NF	16	16	16	16	16	16	16	16
SS	ft(GHz)	84.636	76.978	46.009	40.418	NA	NA	NA	NA
	idsat(uA/um)	2.95E+02	2.93E+02	2.74E+02	2.11E+02	3.07E+02	2.99E+02	2.80E+02	2.05E+02
	vtlin(V)	6.40E-01	6.42E-01	6.62E-01	6.44E-01	6.43E-01	6.48E-01	6.59E-01	6.48E-01
	vtsat(V)	5.76E-01	5.76E-01	5.83E-01	6.01E-01	5.75E-01	5.79E-01	5.83E-01	6.08E-01
	body_effect(V)	1.70E-01	1.70E-01	1.65E-01	1.92E-01	1.84E-01	1.84E-01	1.78E-01	2.03E-01
	ioff(pA/um)	9.28E-01	1.26E+00	3.47E+00	1.36E+00	4.36E-01	4.23E-01	5.58E-01	1.74E-01
SNFP	ft(GHz)	87.6	80.422	50.09	41.87	NA	NA	NA	NA
	idsat(uA/um)	3.26E+02	3.24E+02	3.13E+02	2.34E+02	3.35E+02	3.27E+02	3.14E+02	2.28E+02
	vtlin(V)	6.08E-01	6.08E-01	6.15E-01	6.11E-01	6.11E-01	6.14E-01	6.18E-01	6.16E-01
	vtsat(V)	5.44E-01	5.42E-01	5.39E-01	5.69E-01	5.43E-01	5.45E-01	5.42E-01	5.76E-01
	body_effect(V)	1.64E-01	1.64E-01	1.60E-01	1.86E-01	1.77E-01	1.77E-01	1.71E-01	1.95E-01
	ioff(pA/um)	1.74E+00	2.06E+00	4.22E+00	1.72E+00	9.50E-01	9.16E-01	1.09E+00	2.58E-01
TT	ft(GHz)	91.35	84.45	53.786	43.665	NA	NA	NA	NA
	idsat(uA/um)	3.54E+02	3.53E+02	3.44E+02	2.55E+02	3.63E+02	3.56E+02	3.47E+02	2.52E+02
	vtlin(V)	5.76E-01	5.73E-01	5.72E-01	5.78E-01	5.78E-01	5.79E-01	5.75E-01	5.83E-01
	vtsat(V)	5.12E-01	5.08E-01	4.98E-01	5.36E-01	5.10E-01	5.10E-01	4.99E-01	5.43E-01
	body_effect(V)	1.64E-01	1.64E-01	1.62E-01	1.86E-01	1.76E-01	1.77E-01	1.70E-01	1.96E-01
	ioff(pA/um)	3.31E+00	3.63E+00	5.91E+00	2.01E+00	2.51E+00	2.47E+00	3.18E+00	5.16E-01
FNFP	ft(GHz)	94.319	87.751	57.157	45.441	NA	NA	NA	NA
	idsat(uA/um)	3.82E+02	3.81E+02	3.75E+02	2.77E+02	3.91E+02	3.85E+02	3.82E+02	2.76E+02
	vtlin(V)	5.44E-01	5.40E-01	5.28E-01	5.46E-01	5.45E-01	5.45E-01	5.33E-01	5.50E-01
	vtsat(V)	4.81E-01	4.75E-01	4.55E-01	5.04E-01	4.77E-01	4.75E-01	4.57E-01	5.10E-01
	body_effect(V)	1.64E-01	1.64E-01	1.62E-01	1.86E-01	1.76E-01	1.76E-01	1.70E-01	1.95E-01
	ioff(pA/um)	7.23E+00	7.87E+00	1.23E+01	2.77E+00	6.61E+00	6.79E+00	1.04E+01	1.21E+00
FF	ft(GHz)	96.201	90.603	62.284	46.612	NA	NA	NA	NA
	idsat(uA/um)	4.13E+02	4.14E+02	4.19E+02	3.02E+02	4.19E+02	4.14E+02	4.17E+02	3.00E+02
	vtlin(V)	5.12E-01	5.05E-01	4.83E-01	5.13E-01	5.13E-01	5.12E-01	4.93E-01	5.18E-01
	vtsat(V)	4.49E-01	4.41E-01	4.11E-01	4.71E-01	4.46E-01	4.43E-01	4.17E-01	4.80E-01
	body_effect(V)	1.58E-01	1.58E-01	1.56E-01	1.80E-01	1.69E-01	1.69E-01	1.63E-01	1.88E-01
	ioff(pA/um)	1.69E+01	1.90E+01	3.47E+01	5.45E+00	1.50E+01	1.62E+01	3.06E+01	2.67E+00

Table 2-1 RF & Logic Corner Table at 25 °C

Temp=125C		RF Model				Logic Model			
Device	LF (um)	0.12	0.12	0.12	0.18	0.12	0.12	0.12	0.18
	WF (um)	7.2	3.6	0.9	2.4	7.2	3.6	0.9	2.4
	NF	16	16	16	16	16	16	16	16
SS	ft(GHz)	66.035	59.719	34.865	30.266	NA	NA	NA	NA
	idsat(uA/um)	2.52E+02	2.52E+02	2.38E+02	1.77E+02	2.67E+02	2.60E+02	2.48E+02	1.80E+02
	vtlin(V)	5.77E-01	5.75E-01	5.82E-01	5.76E-01	5.76E-01	5.81E-01	5.90E-01	5.82E-01
	vtsat(V)	5.00E-01	4.98E-01	4.96E-01	5.23E-01	5.00E-01	5.03E-01	5.05E-01	5.34E-01
	body_effect(V)	1.86E-01	1.87E-01	1.82E-01	2.09E-01	2.01E-01	2.01E-01	1.95E-01	2.19E-01
	ioff(pA/um)	9.25E+01	8.98E+01	7.57E+01	2.64E+01	8.76E+01	7.99E+01	6.91E+01	2.11E+01
SNFP	ft(GHz)	68.269	62.345	38.052	31.458	NA	NA	NA	NA
	idsat(uA/um)	2.75E+02	2.76E+02	2.69E+02	1.95E+02	2.88E+02	2.82E+02	2.71E+02	1.98E+02
	vtlin(V)	5.44E-01	5.40E-01	5.35E-01	5.42E-01	5.43E-01	5.46E-01	5.48E-01	5.49E-01
	vtsat(V)	4.68E-01	4.63E-01	4.51E-01	4.90E-01	4.67E-01	4.69E-01	4.63E-01	5.01E-01
	body_effect(V)	1.80E-01	1.80E-01	1.77E-01	2.02E-01	1.94E-01	1.94E-01	1.88E-01	2.12E-01
	ioff(pA/um)	1.74E+02	1.78E+02	1.87E+02	5.04E+01	1.66E+02	1.58E+02	1.63E+02	4.05E+01
TT	ft(GHz)	70.562	64.91	40.631	32.537	NA	NA	NA	NA
	idsat(uA/um)	2.96E+02	2.97E+02	2.92E+02	2.10E+02	3.08E+02	3.03E+02	2.94E+02	2.16E+02
	vtlin(V)	5.12E-01	5.05E-01	4.91E-01	5.09E-01	5.10E-01	5.11E-01	5.05E-01	5.15E-01
	vtsat(V)	4.35E-01	4.29E-01	4.10E-01	4.57E-01	4.33E-01	4.33E-01	4.20E-01	4.68E-01
	body_effect(V)	1.81E-01	1.81E-01	1.79E-01	2.03E-01	1.93E-01	1.94E-01	1.88E-01	2.12E-01
	ioff(pA/um)	3.59E+02	3.79E+02	4.66E+02	1.05E+02	3.53E+02	3.48E+02	4.30E+02	8.71E+01
FNFP	ft(GHz)	73.27	67.787	43.234	34.217	NA	NA	NA	NA
	idsat(uA/um)	3.19E+02	3.21E+02	3.16E+02	2.28E+02	3.29E+02	3.24E+02	3.18E+02	2.33E+02
	vtlin(V)	4.80E-01	4.72E-01	4.48E-01	4.76E-01	4.76E-01	4.76E-01	4.62E-01	4.82E-01
	vtsat(V)	4.03E-01	3.95E-01	3.66E-01	4.24E-01	4.00E-01	3.98E-01	3.78E-01	4.35E-01
	body_effect(V)	1.81E-01	1.82E-01	1.79E-01	2.03E-01	1.94E-01	1.94E-01	1.88E-01	2.12E-01
	ioff(pA/um)	7.24E+02	8.03E+02	1.24E+03	2.20E+02	7.37E+02	7.56E+02	1.11E+03	1.85E+02
FF	ft(GHz)	74.86	70.159	47.422	35.374	NA	NA	NA	NA
	idsat(uA/um)	3.43E+02	3.47E+02	3.53E+02	2.48E+02	3.50E+02	3.45E+02	3.42E+02	2.51E+02
	vtlin(V)	4.46E-01	4.36E-01	4.02E-01	4.43E-01	4.44E-01	4.42E-01	4.21E-01	4.50E-01
	vtsat(V)	3.71E-01	3.61E-01	3.22E-01	3.92E-01	3.69E-01	3.66E-01	3.38E-01	4.04E-01
	body_effect(V)	1.75E-01	1.75E-01	1.73E-01	1.97E-01	1.86E-01	1.87E-01	1.81E-01	2.06E-01
	ioff(pA/um)	1.39E+03	1.62E+03	3.15E+03	4.37E+02	1.39E+03	1.48E+03	2.55E+03	3.54E+02

Table 2-2 RF & Logic Corner Table at 125 °C

Temp=-50C		RF Model				Logic Model			
Device	LF (um)	0.12	0.12	0.12	0.18	0.12	0.12	0.12	0.18
	WF (um)	7.2	3.6	0.9	2.4	7.2	3.6	0.9	2.4
	NF	16	16	16	16	16	16	16	16
SS	ft(GHz)	97.171	88.677	53.384	45.345	NA	NA	NA	NA
	idsat(uA/um)	3.21E+02	3.13E+02	2.84E+02	2.24E+02	3.39E+02	3.29E+02	3.05E+02	2.27E+02
	vtlin(V)	7.00E-01	7.06E-01	7.36E-01	7.08E-01	7.06E-01	7.11E-01	7.23E-01	7.10E-01
	vtsat(V)	6.45E-01	6.47E-01	6.62E-01	6.71E-01	6.43E-01	6.46E-01	6.52E-01	6.74E-01
	body_effect(V)	1.59E-01	1.59E-01	1.53E-01	1.81E-01	1.72E-01	1.73E-01	1.65E-01	1.91E-01
	ioff(pA/um)	4.91E-01	8.85E-01	3.25E+00	1.28E+00	3.74E-02	7.42E-02	2.97E-01	1.11E-01
SNFP	ft(GHz)	100.86	92.898	58.097	46.793	NA	NA	NA	NA
	idsat(uA/um)	3.56E+02	3.49E+02	3.27E+02	2.49E+02	3.70E+02	3.62E+02	3.44E+02	2.54E+02
	vtlin(V)	6.68E-01	6.71E-01	6.90E-01	6.76E-01	6.74E-01	6.78E-01	6.82E-01	6.78E-01
	vtsat(V)	6.13E-01	6.14E-01	6.18E-01	6.39E-01	6.12E-01	6.14E-01	6.11E-01	6.43E-01
	body_effect(V)	1.52E-01	1.52E-01	1.48E-01	1.74E-01	1.65E-01	1.65E-01	1.58E-01	1.84E-01
	ioff(pA/um)	7.26E-01	1.12E+00	3.48E+00	1.51E+00	3.78E-02	7.47E-02	2.97E-01	1.11E-01
TT	ft(GHz)	106.04	98.305	62.658	48.943	NA	NA	NA	NA
	idsat(uA/um)	3.90E+02	3.84E+02	3.64E+02	2.74E+02	4.03E+02	3.95E+02	3.86E+02	2.82E+02
	vtlin(V)	6.36E-01	6.37E-01	6.46E-01	6.43E-01	6.41E-01	6.43E-01	6.39E-01	6.45E-01
	vtsat(V)	5.81E-01	5.80E-01	5.78E-01	6.07E-01	5.78E-01	5.78E-01	5.68E-01	6.10E-01
	body_effect(V)	1.52E-01	1.52E-01	1.49E-01	1.74E-01	1.64E-01	1.65E-01	1.58E-01	1.83E-01
	ioff(pA/um)	7.29E-01	1.12E+00	3.48E+00	1.51E+00	4.04E-02	7.72E-02	3.00E-01	1.11E-01
FNFP	ft(GHz)	109.35	102.04	66.573	50.629	NA	NA	NA	NA
	idsat(uA/um)	4.22E+02	4.17E+02	4.00E+02	2.98E+02	4.35E+02	4.29E+02	4.27E+02	3.11E+02
	vtlin(V)	6.05E-01	6.04E-01	6.03E-01	6.11E-01	6.09E-01	6.09E-01	5.98E-01	6.12E-01
	vtsat(V)	5.50E-01	5.47E-01	5.35E-01	5.74E-01	5.45E-01	5.44E-01	5.26E-01	5.77E-01
	body_effect(V)	1.52E-01	1.53E-01	1.49E-01	1.74E-01	1.64E-01	1.64E-01	1.57E-01	1.83E-01
	ioff(pA/um)	7.37E-01	1.13E+00	3.49E+00	1.51E+00	4.88E-02	8.61E-02	3.15E-01	1.12E-01
FF	ft(GHz)	111.62	105.39	72.321	51.626	NA	NA	NA	NA
	idsat(uA/um)	4.58E+02	4.55E+02	4.50E+02	3.25E+02	4.67E+02	4.62E+02	4.69E+02	3.39E+02
	vtlin(V)	5.73E-01	5.70E-01	5.58E-01	5.78E-01	5.78E-01	5.76E-01	5.58E-01	5.82E-01
	vtsat(V)	5.19E-01	5.13E-01	4.91E-01	5.42E-01	5.15E-01	5.12E-01	4.87E-01	5.46E-01
	body_effect(V)	1.45E-01	1.46E-01	1.43E-01	1.67E-01	1.57E-01	1.57E-01	1.50E-01	1.75E-01
	ioff(pA/um)	1.57E+00	1.96E+00	4.32E+00	2.32E+00	7.15E-02	1.11E-01	3.78E-01	1.15E-01

Table 2-3 RF & Logic Corner Table at -50 °C

2.1.2 Summary Table

In the following fitting plots & summary table, the parasitic resistances due to metal interconnection from pad to DUT (Device Under Test) are taken into account. Since RF DUT generally use larger total gate width and has larger current, these parasitic resistances will cause voltage drops. Therefore the extracted I_{dsat} is smaller than the real I_{dsat} for the DUT without the parasitic resistance.

Device						Vtlin(V)		Vtsat(V)		Idsat(uA/um)		Ioff(pA/um)		ft(GHz)	
Name	LF(um)	WF(um)	NF	M	con	Measure	Model	Measure	Model	Measure	Model	Measure	Model	Measure	Model
NFET551	0.12	1.8	16	4	1	0.613	0.612	0.522	0.514	346.39	341.62	3.66	3.974	73.642	75.112
NFET552	0.12	2.4	16	3	1	0.616	0.614	0.525	0.516	346.2	338.84	3.25	3.635	82.579	79.58
NFET553	0.12	3.6	16	2	1	0.616	0.617	0.525	0.520	337.06	333.59	3.59	3.169	84.479	83.477
NFET554	0.12	7.2	16	1	1	0.613	0.621	0.524	0.527	319.75	322.34	3.39	2.624	85.293	84.554
NFET555	0.12	0.9	16	6	1	0.608	0.609	0.532	0.532	329.99	331.88	3.27	4.466	56.857	57.291
NFET556	0.12	1.8	2	18	1	0.621	0.612	0.535	0.514	331.17	345.61	4.27	4.622	64.986	64.873
NFET558	0.12	1.8	5	13	1	0.620	0.612	0.532	0.514	333.47	343.89	3.29	4.178	71.683	71.778
NFET559	0.12	1.8	8	8	1	0.617	0.612	0.527	0.514	343.52	342.91	3.27	4.067	73.578	73.55
NFET560	0.12	1.8	11	6	1	0.616	0.612	0.526	0.514	344.74	342.15	3.15	4.016	74.583	74.335
NFET561	0.12	1.8	22	3	1	0.611	0.613	0.522	0.514	347.53	340.96	3.99	3.949	75.336	75.17
NFET563	0.15	2.4	16	3	1	0.602	0.603	0.547	0.540	277.57	280.85	2.01	2.238	56.892	58.847
NFET564	0.18	2.4	16	3	1	0.586	0.594	0.541	0.543	246.03	246.06	1.71	1.997	44.41	45.755
NFET565	0.24	3.6	16	2	1	0.571	0.581	0.535	0.540	198.15	198.07	1.61	1.471	31.117	30.574
NFET566	0.36	3.6	16	2	1	0.552	0.554	0.524	0.523	147.01	150.61	1.87	1.43	17.437	16.764
NFET570	0.12	3.6	4	8	1	0.623	0.617	0.533	0.520	328.09	337.68	3.08	3.308	78.447	80.814
NFET571	0.12	3.6	11	4	1	0.616	0.617	0.526	0.520	335.31	333.91	3.35	3.19	81.716	82.822
NFET572	0.12	3.6	22	2	1	0.614	0.617	0.524	0.520	337.46	331.32	3.39	3.157	83.135	82.25
NFET573	0.15	7.2	4	4	1	0.608	0.608	0.555	0.543	257.79	273.77	1.53	1.567	58.906	60.23

Table 2-4 Summary Table for Twin Well RF NMOS

Measure: Measurement result.

Model: Simulation result

2.1.3 Simulation results for WF, LF and NF trends

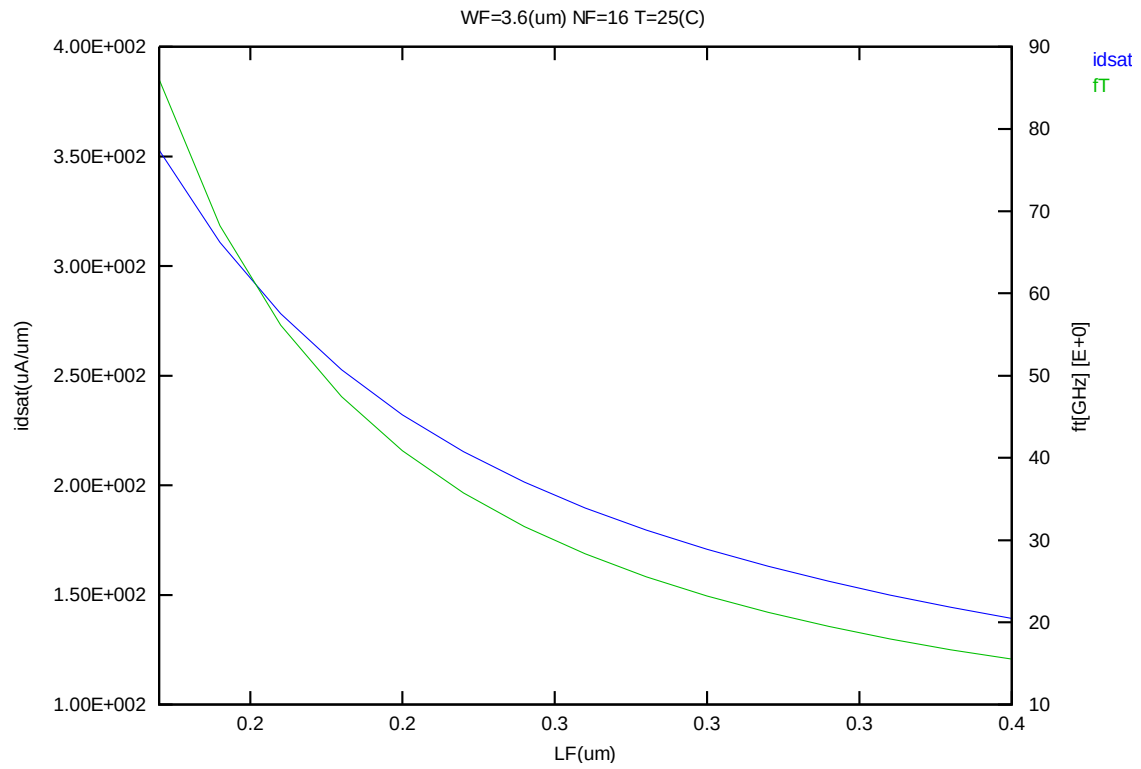


Figure 2-1 fT and Idsat vs. channel length

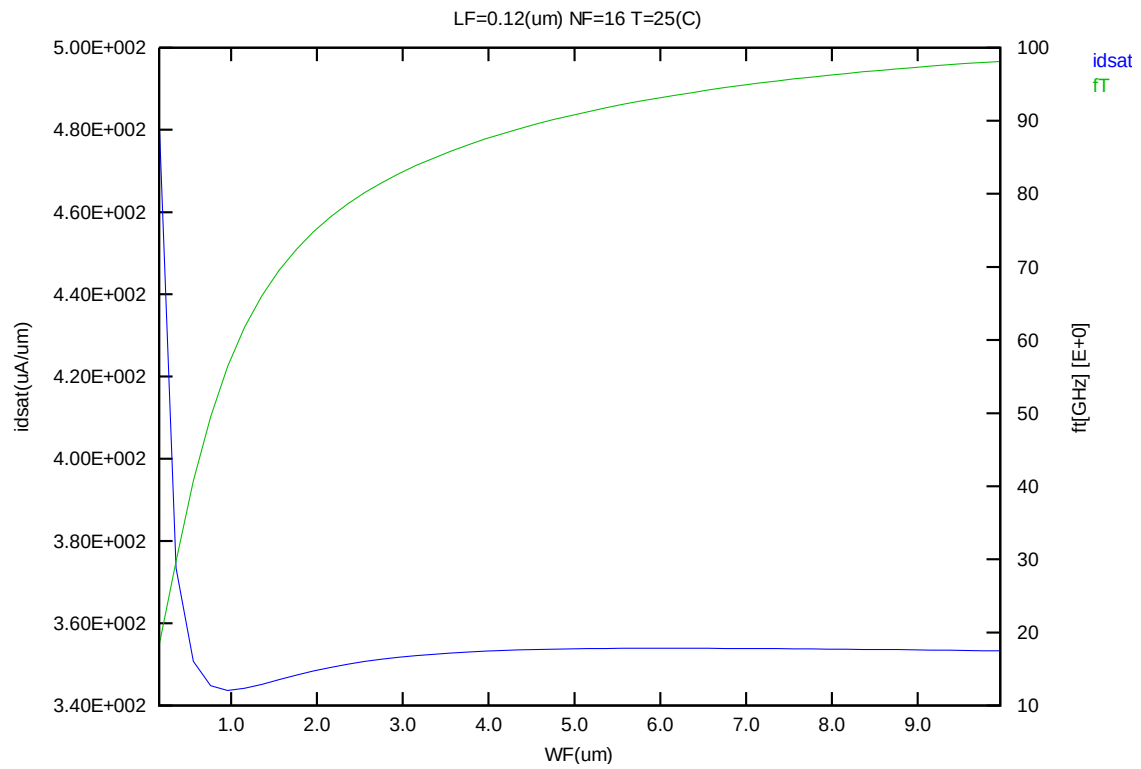


Figure 2-2 fT and Idsat vs. channel width

2.1.4 DC & RF Fitting curves for individual devices

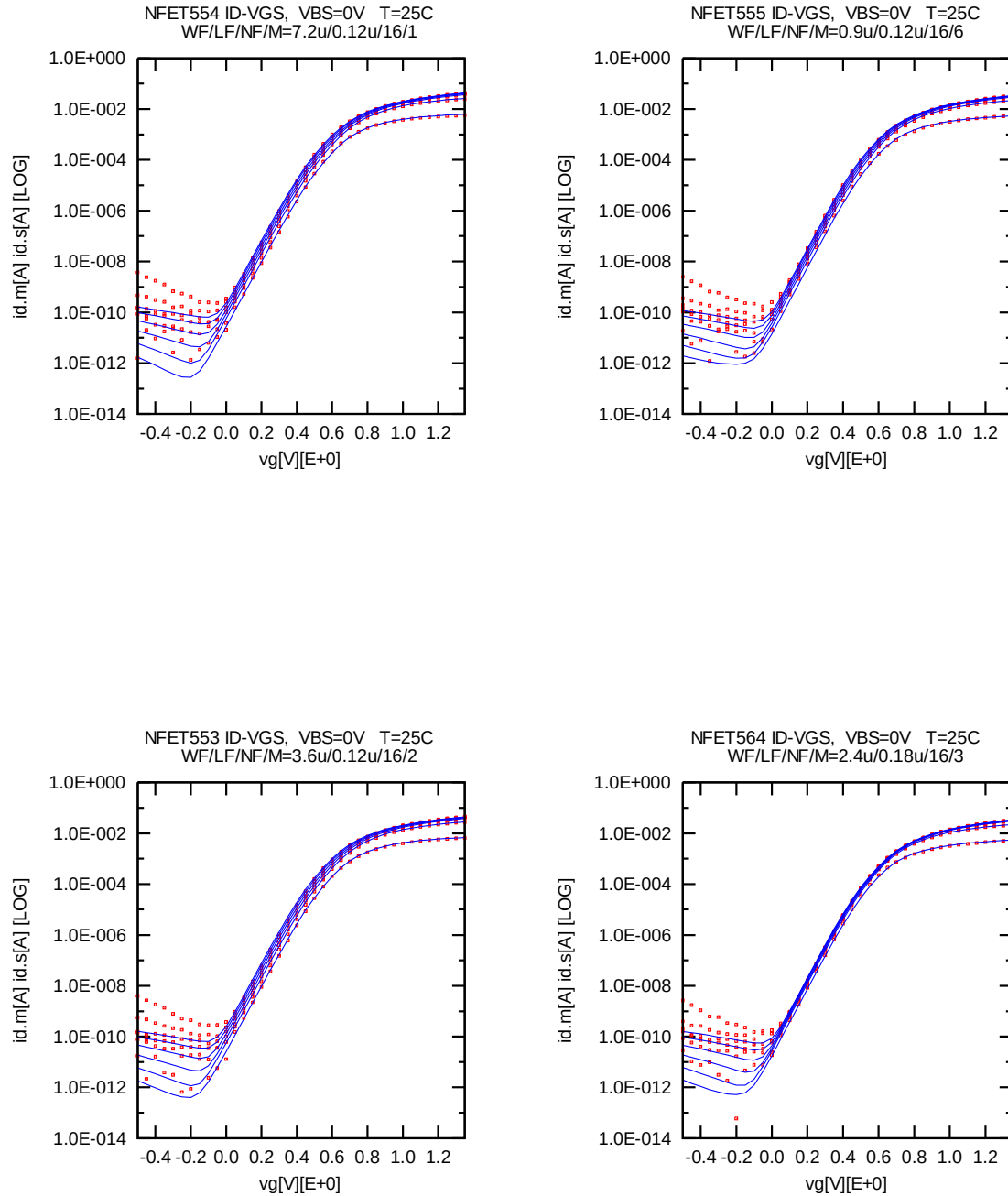


Figure 2-3 Id-Vg (25°C)

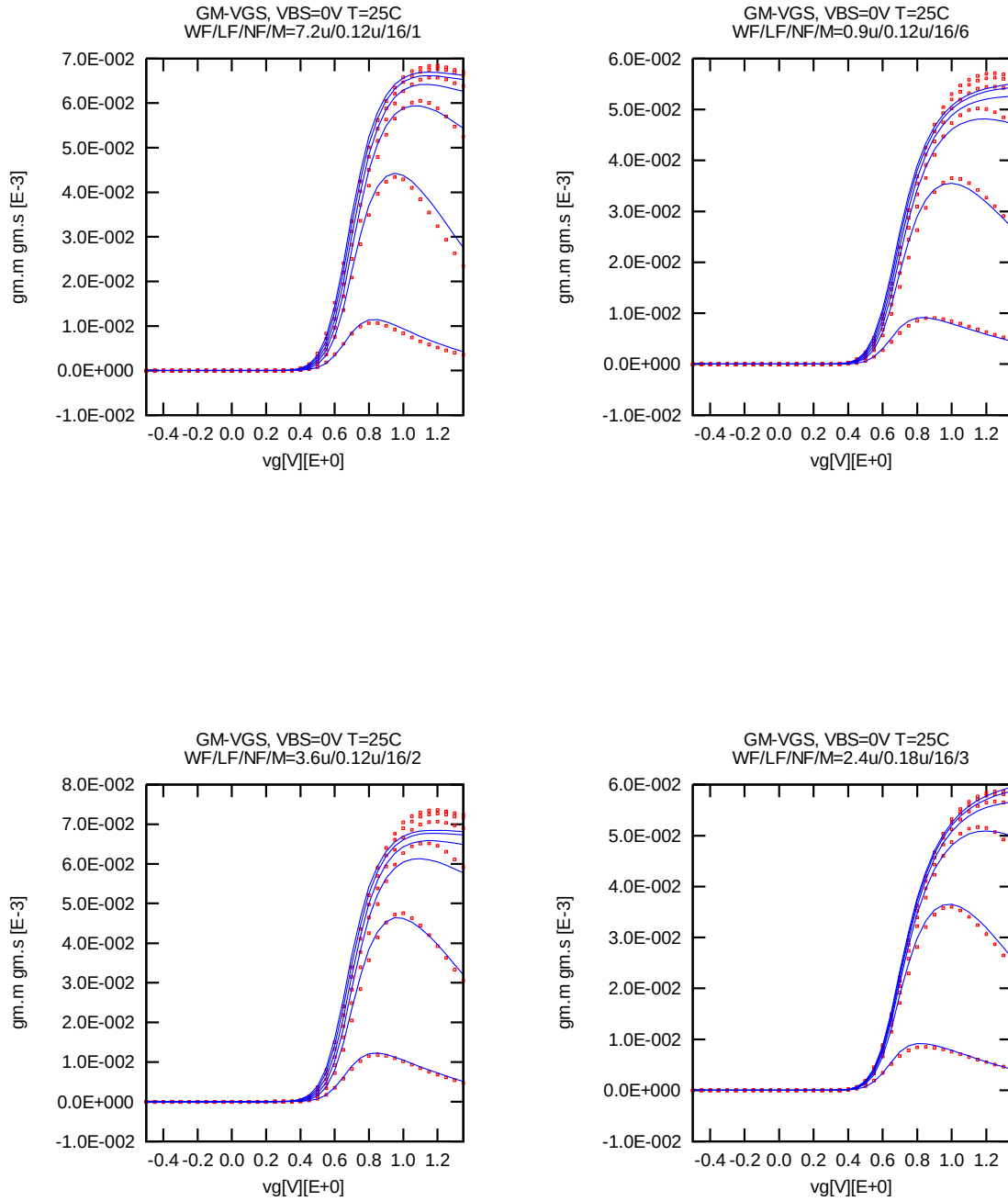


Figure 2-4 Gm-Vg (25°C)

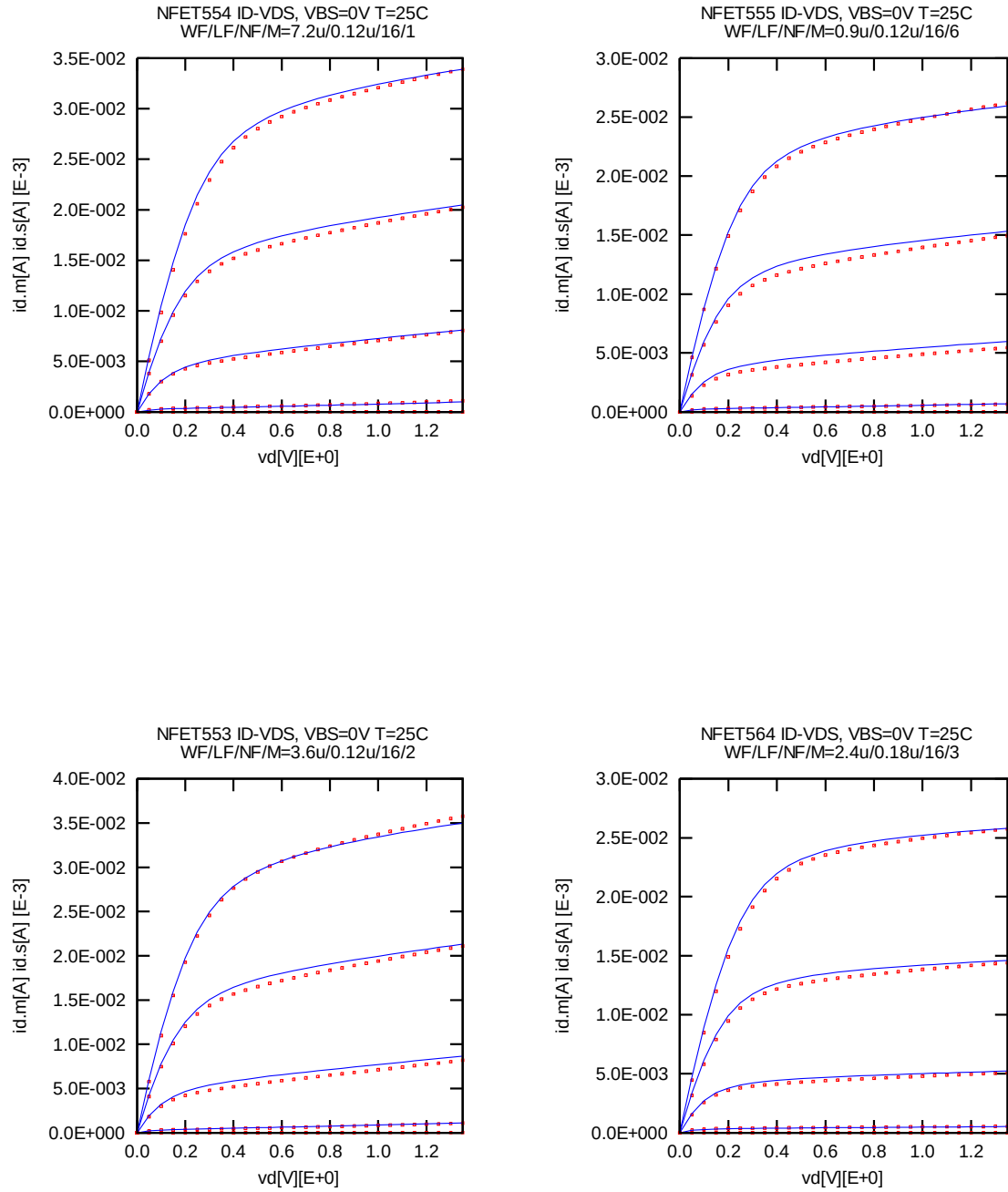


Figure 2-5 Id-Vd (25°C)

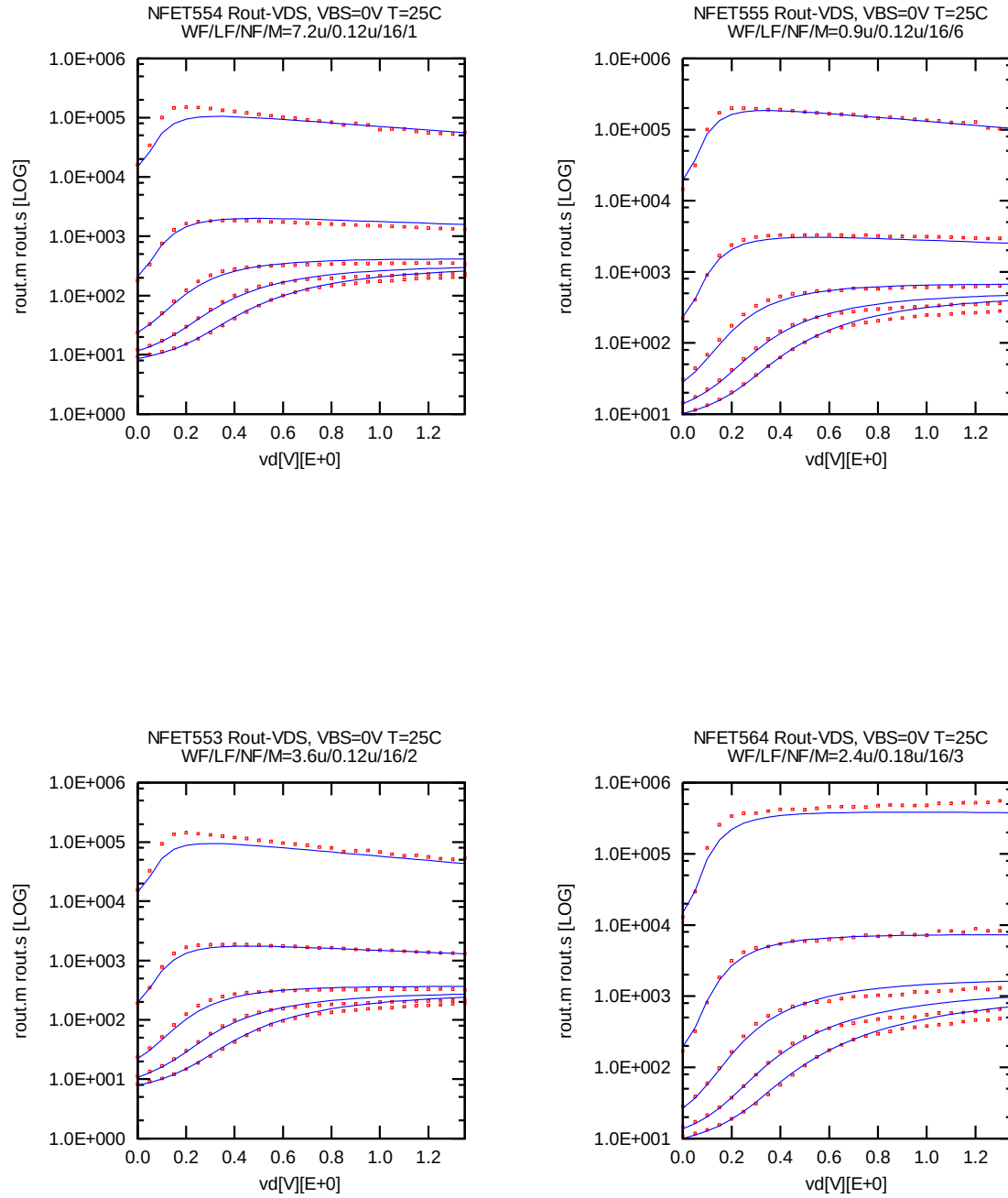


Figure 2-6 Rout-Vd (25°C)

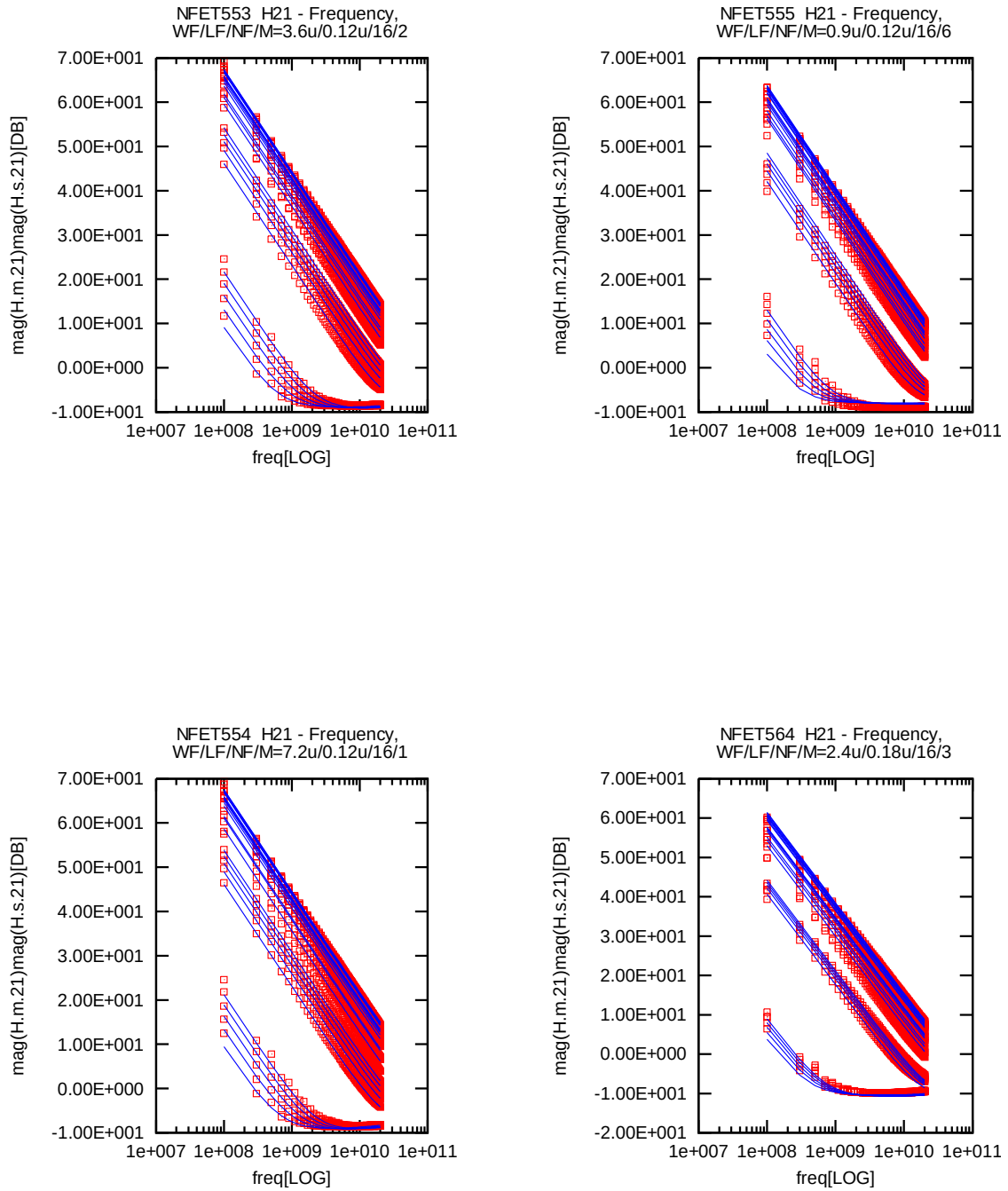


Figure 2-7 Mag(H21)-Freq (25°C)

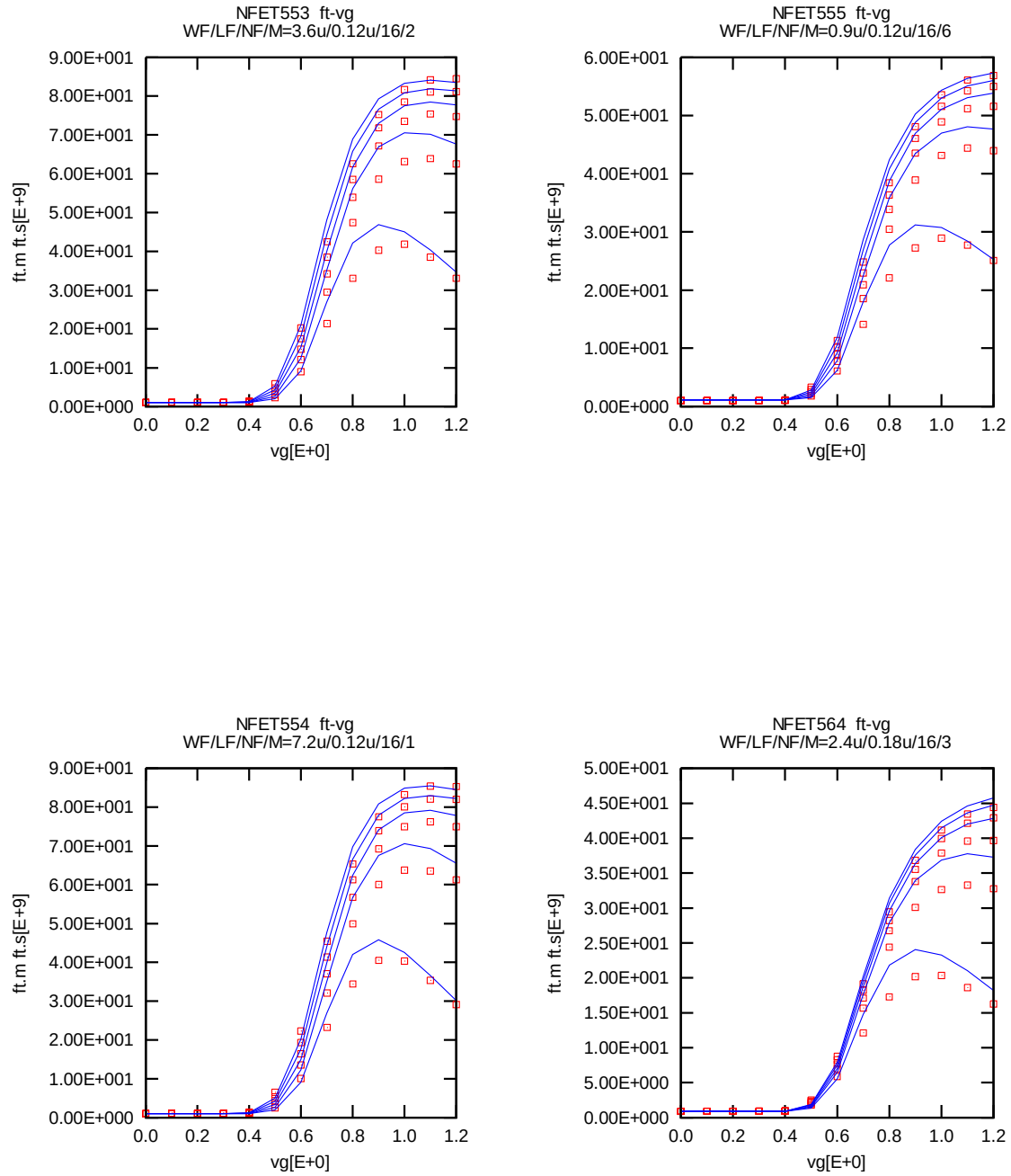


Figure 2-8 fT-Vg (25°C)

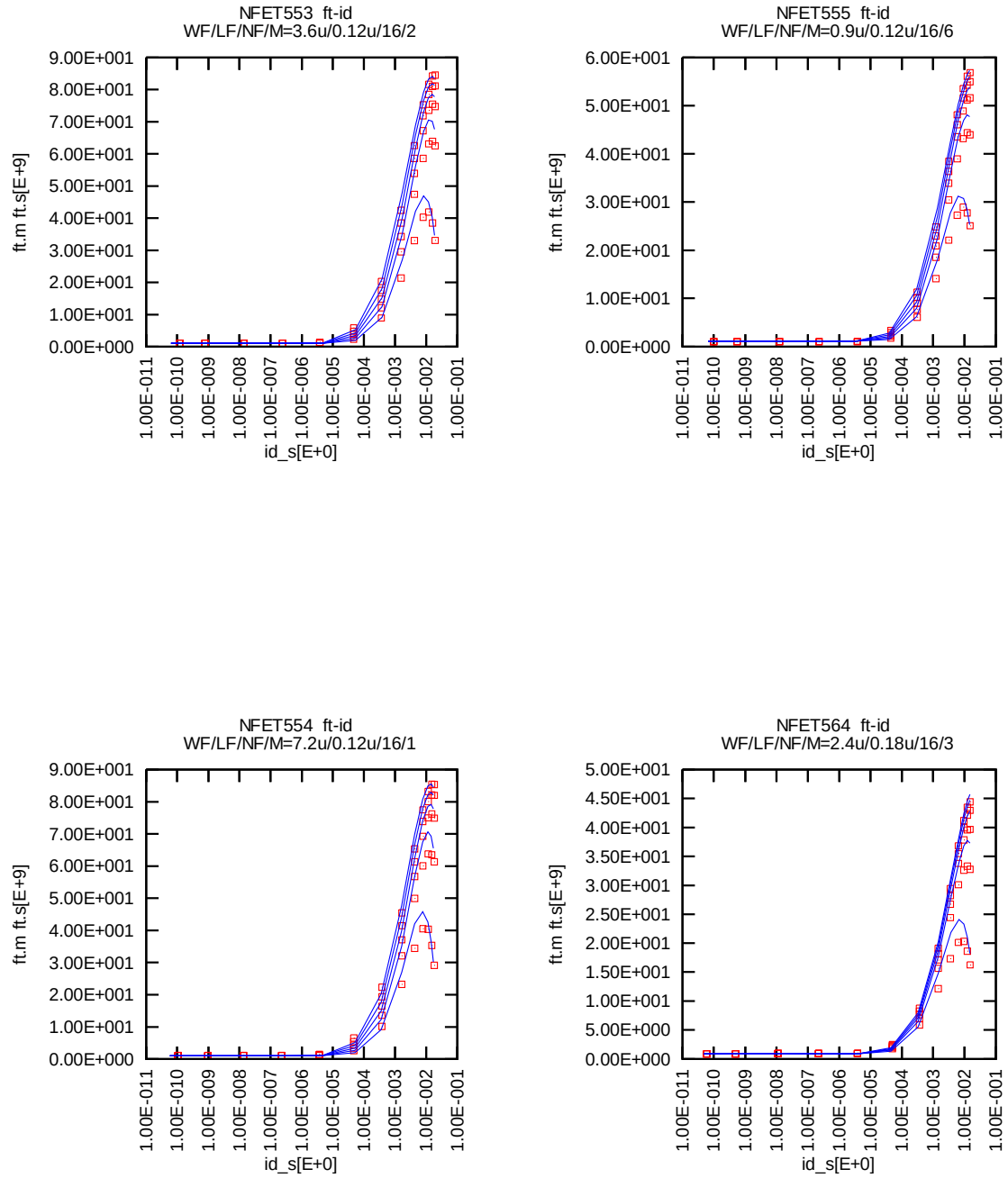
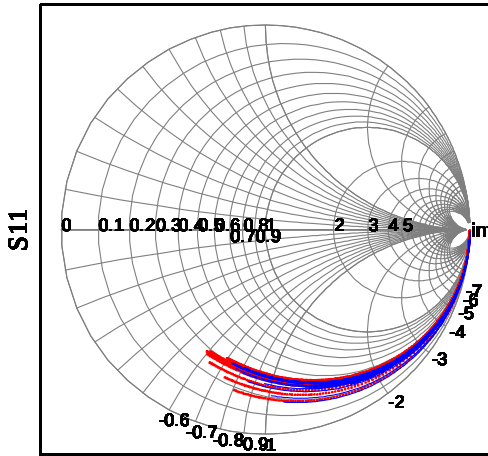
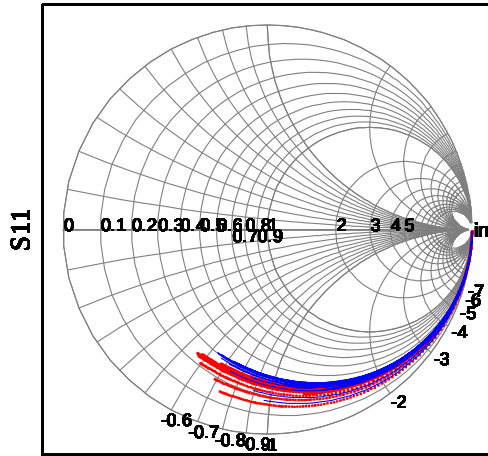


Figure 2-9 fT-Id (25°C)

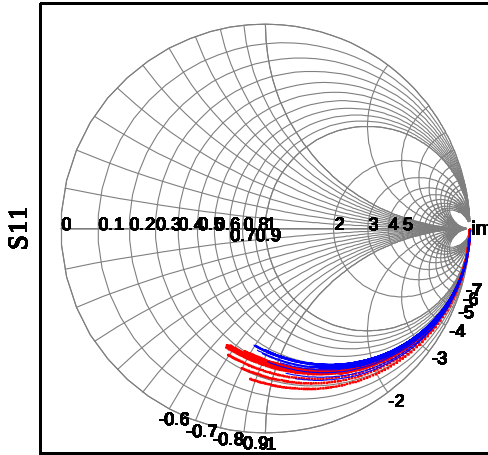
NFET553 S11 WF/LF/NF/M=3.6u/0.12u/16/2



NFET555 S11 WF/LF/NF/M=0.9u/0.12u/16/6



NFET554 S11 WF/LF/NF/M=7.2u/0.12u/16/1



NFET564 S11 WF/LF/NF/M=2.4u/0.18u/16/3

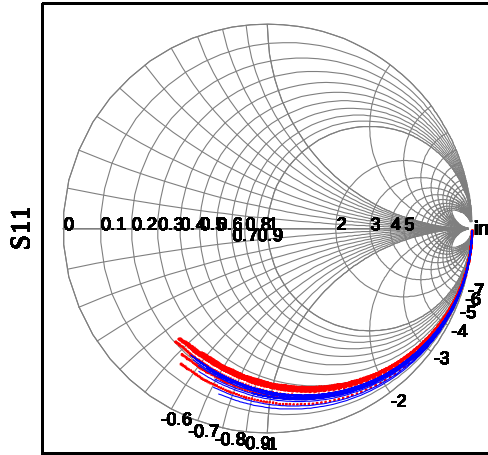


Figure 2-10 S11 (25°C)

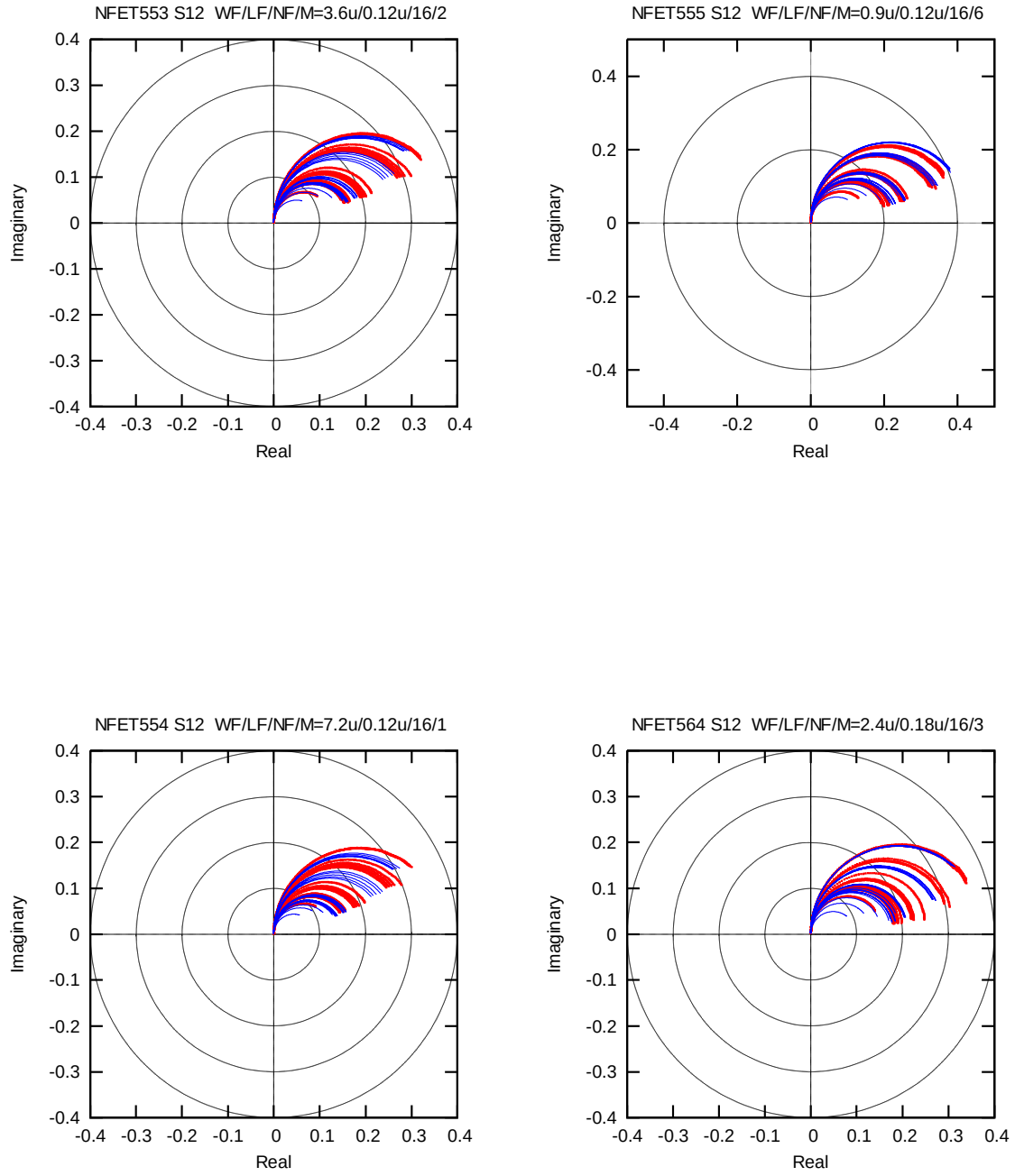


Figure 2-11 S12 (25°C)

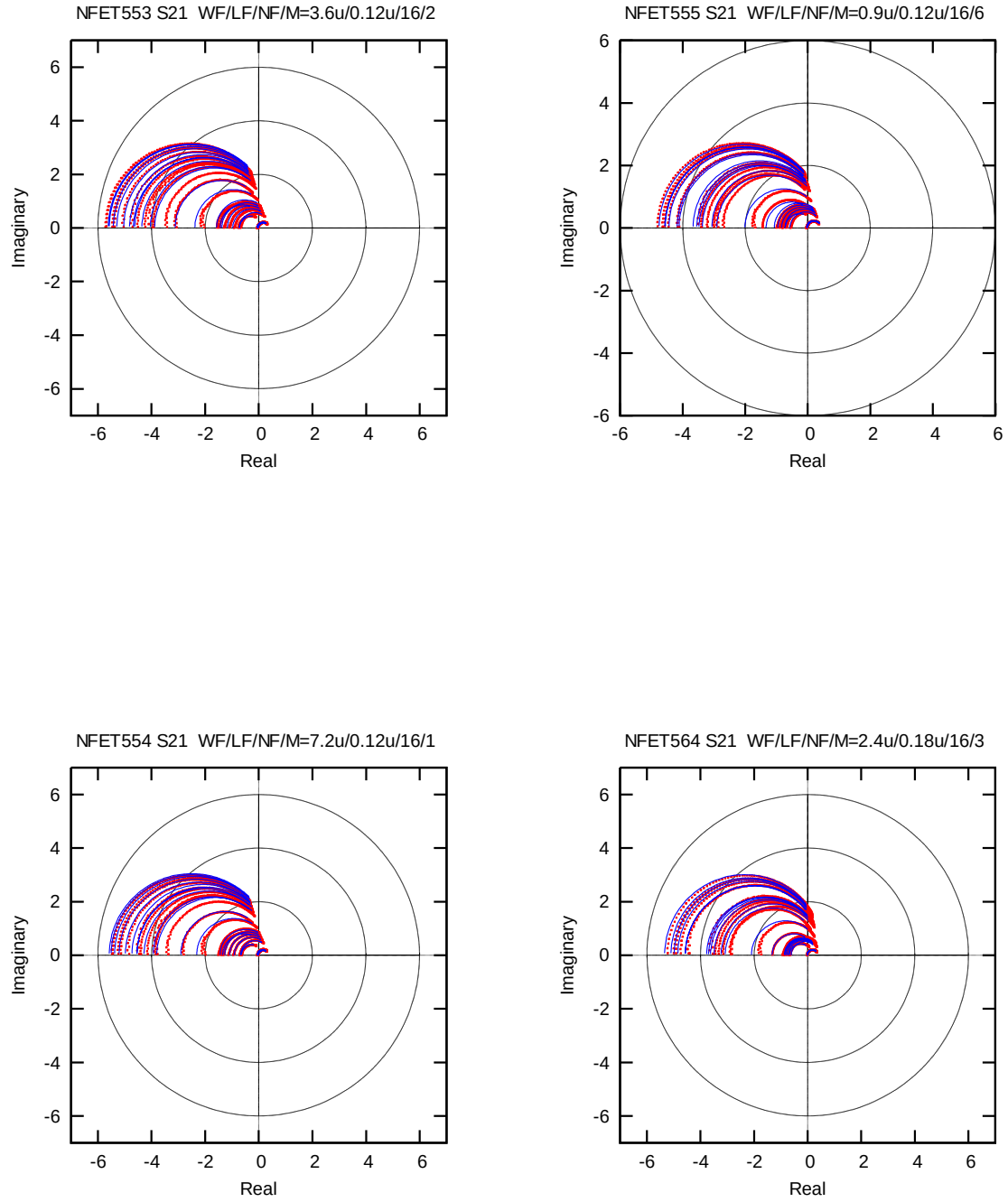
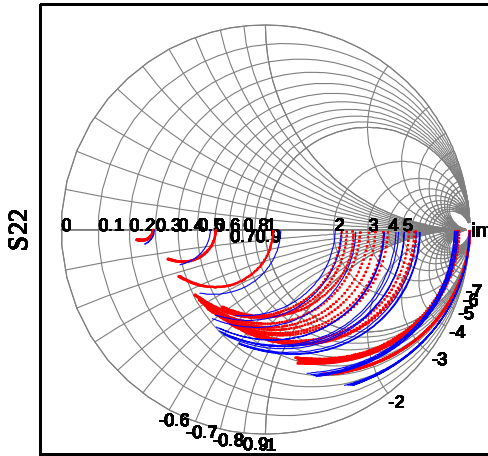
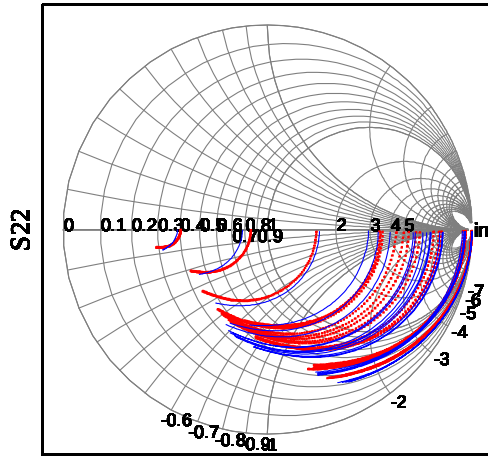


Figure 2-12 S21 (25°C)

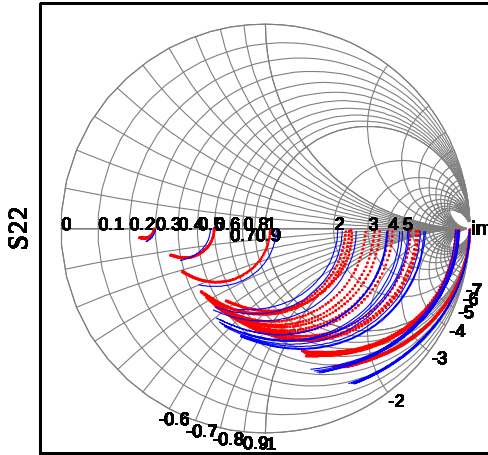
NFET553 S22 WF/LF/NF/M=3.6u/0.12u/16/2



NFET555 S22 WF/LF/NF/M=0.9u/0.12u/16/6



NFET554 S22 WF/LF/NF/M=7.2u/0.12u/16/1



NFET564 S22 WF/LF/NF/M=2.4u/0.18u/16/3

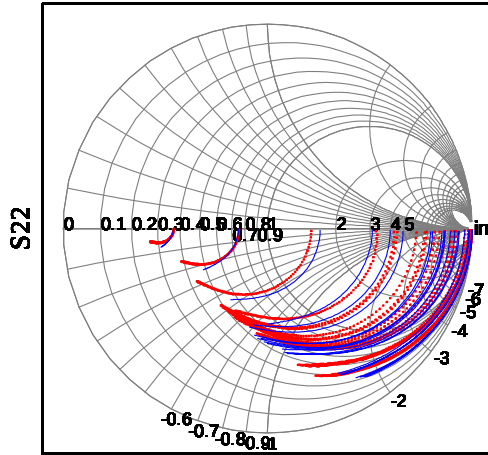


Figure 2-13 S22 (25°C)

2.2 PMOS Model

2.2.1 Corner Table

The tables below summarize the DC & RF simulation results for different corner cases for some selected devices and compared with Logic/MM model.

The biasing conditions for the parameters are:

ft : Cut-off frequency extracted @ Ft_max:Vd=-1.2V, sweep Vg
vth : Linear Vth extracted @ Vd=-0.05V @
Id=-70nA*WF*0.9*NF/(LF*0.9+0.012um)
vtsat : Saturation Vth extracted @ Vd=-1.2V using fixed current
Id=-70nA*WF*0.9*NF/(LF*0.9+0.012um)
body effect : vtsat shift for Vbody =0 ~ 1.2V
Idsat : Drain current extracted @ Vd=Vg=-1.2V
Idlin : Drain current extracted @ Vd=-0.05, Vg=-1.2V
Ioff : Drain current extracted @ Vg=0V, Vd=-1.2V

Temp=25C		RF Model				Logic Model			
Device	LF (um)	0.24	0.12	0.12	0.12	0.24	0.12	0.12	0.12
	WF (um)	9.6	9.6	2.4	1.6	9.6	9.6	2.4	1.6
	NF	22	16	4	16	22	16	4	16
SS	ft(GHz)	10.486	35.861	28.677	27.702	NA	NA	NA	NA
	idsat(uA/um)	-4.50E+01	-1.01E+02	-9.74E+01	-9.72E+01	-4.92E+01	-1.03E+02	-9.36E+01	-9.00E+01
	vtlin(V)	-5.76E-01	-5.86E-01	-5.91E-01	-5.93E-01	-5.72E-01	-5.85E-01	-5.96E-01	-6.02E-01
	vtsat(V)	-5.44E-01	-5.29E-01	-5.35E-01	-5.38E-01	-5.42E-01	-5.23E-01	-5.37E-01	-5.44E-01
	body_effect(V)	-2.96E-01	-2.47E-01	-2.46E-01	-2.44E-01	-2.95E-01	-2.48E-01	-2.46E-01	-2.45E-01
	ioff(pA/um)	-1.06E-01	-5.40E-01	-4.99E-01	-4.78E-01	-8.39E-02	-6.07E-01	-4.34E-01	-3.98E-01
SNFP	ft(GHz)	11.021	41.312	35.221	34.538	NA	NA	NA	NA
	idsat(uA/um)	-5.74E+01	-1.44E+02	-1.43E+02	-1.46E+02	-5.97E+01	-1.42E+02	-1.34E+02	-1.31E+02
	vtlin(V)	-4.98E-01	-4.92E-01	-4.87E-01	-4.82E-01	-4.94E-01	-4.91E-01	-4.92E-01	-4.92E-01
	vtsat(V)	-4.68E-01	-4.34E-01	-4.30E-01	-4.27E-01	-4.66E-01	-4.31E-01	-4.34E-01	-4.35E-01
	body_effect(V)	-2.87E-01	-2.35E-01	-2.33E-01	-2.33E-01	-2.86E-01	-2.35E-01	-2.34E-01	-2.33E-01
	ioff(pA/um)	-5.94E-01	-6.79E+00	-6.53E+00	-6.51E+00	-5.38E-01	-7.42E+00	-5.85E+00	-5.25E+00
TT	ft(GHz)	10.877	39.287	32.944	32.188	NA	NA	NA	NA
	idsat(uA/um)	-5.38E+01	-1.30E+02	-1.29E+02	-1.30E+02	-5.57E+01	-1.29E+02	-1.20E+02	-1.17E+02
	vtlin(V)	-5.24E-01	-5.23E-01	-5.21E-01	-5.19E-01	-5.20E-01	-5.22E-01	-5.26E-01	-5.28E-01
	vtsat(V)	-4.93E-01	-4.64E-01	-4.64E-01	-4.63E-01	-4.91E-01	-4.61E-01	-4.68E-01	-4.71E-01
	body_effect(V)	-2.87E-01	-2.36E-01	-2.34E-01	-2.33E-01	-2.86E-01	-2.36E-01	-2.35E-01	-2.34E-01
	ioff(pA/um)	-3.35E-01	-2.95E+00	-2.62E+00	-2.47E+00	-2.64E-01	-3.20E+00	-2.30E+00	-1.96E+00
FNFP	ft(GHz)	10.69	36.88	30.366	29.528	NA	NA	NA	NA
	idsat(uA/um)	-5.01E+01	-1.16E+02	-1.13E+02	-1.14E+02	-5.30E+01	-1.16E+02	-1.07E+02	-1.03E+02
	vtlin(V)	-5.50E-01	-5.56E-01	-5.57E-01	-5.57E-01	-5.46E-01	-5.54E-01	-5.62E-01	-5.66E-01
	vtsat(V)	-5.19E-01	-4.98E-01	-5.01E-01	-5.02E-01	-5.18E-01	-4.93E-01	-5.04E-01	-5.09E-01
	body_effect(V)	-2.87E-01	-2.38E-01	-2.36E-01	-2.35E-01	-2.86E-01	-2.38E-01	-2.37E-01	-2.36E-01
	ioff(pA/um)	-2.13E-01	-1.15E+00	-1.02E+00	-9.65E-01	-1.35E-01	-1.23E+00	-8.60E-01	-7.43E-01
FF	ft(GHz)	11.158	41.439	36.269	35.771	NA	NA	NA	NA
	idsat(uA/um)	-6.32E+01	-1.60E+02	-1.61E+02	-1.65E+02	-6.33E+01	-1.55E+02	-1.48E+02	-1.45E+02
	vtlin(V)	-4.74E-01	-4.63E-01	-4.54E-01	-4.48E-01	-4.69E-01	-4.62E-01	-4.60E-01	-4.57E-01
	vtsat(V)	-4.44E-01	-4.06E-01	-3.99E-01	-3.93E-01	-4.43E-01	-4.03E-01	-4.03E-01	-4.02E-01
	body_effect(V)	-2.79E-01	-2.28E-01	-2.26E-01	-2.25E-01	-2.78E-01	-2.28E-01	-2.27E-01	-2.26E-01
	ioff(pA/um)	-1.23E+00	-1.34E+01	-1.44E+01	-1.53E+01	-9.88E-01	-1.43E+01	-1.26E+01	-1.20E+01

Table 2-5 RF & Logic Corner Table at 25 °C

Temp=125C		RF Model				Logic Model			
Device	LF (um)	0.24	0.12	0.12	0.12	0.24	0.12	0.12	0.12
	WF (um)	9.6	9.6	2.4	1.6	9.6	9.6	2.4	1.6
	NF	22	16	4	16	22	16	4	16
SS	ft(GHz)	10.089	33.554	26.818	25.92	NA	NA	NA	NA
	idsat(uA/um)	-4.64E+01	-1.02E+02	-9.84E+01	-9.83E+01	-5.14E+01	-1.04E+02	-9.49E+01	-9.12E+01
	vtlin(V)	-5.03E-01	-5.11E-01	-5.16E-01	-5.18E-01	-4.98E-01	-5.09E-01	-5.21E-01	-5.28E-01
	vtsat(V)	-4.64E-01	-4.47E-01	-4.54E-01	-4.57E-01	-4.61E-01	-4.41E-01	-4.56E-01	-4.64E-01
	body_effect(V)	-3.11E-01	-2.63E-01	-2.61E-01	-2.60E-01	-3.10E-01	-2.63E-01	-2.61E-01	-2.60E-01
	ioff(pA/um)	-1.44E+01	-8.23E+01	-6.15E+01	-5.23E+01	-1.56E+01	-9.30E+01	-5.97E+01	-4.66E+01
SNFP	ft(GHz)	10.242	37.877	32.15	31.46	NA	NA	NA	NA
	idsat(uA/um)	-5.67E+01	-1.40E+02	-1.39E+02	-1.40E+02	-5.89E+01	-1.38E+02	-1.29E+02	-1.26E+02
	vtlin(V)	-4.23E-01	-4.15E-01	-4.10E-01	-4.05E-01	-4.18E-01	-4.13E-01	-4.15E-01	-4.15E-01
	vtsat(V)	-3.86E-01	-3.50E-01	-3.47E-01	-3.44E-01	-3.84E-01	-3.47E-01	-3.51E-01	-3.52E-01
	body_effect(V)	-3.03E-01	-2.51E-01	-2.50E-01	-2.49E-01	-3.02E-01	-2.52E-01	-2.51E-01	-2.50E-01
	ioff(pA/um)	-7.92E+01	-6.03E+02	-5.77E+02	-5.74E+02	-8.35E+01	-6.49E+02	-5.35E+02	-4.90E+02
TT	ft(GHz)	10.268	36.398	30.423	29.687	NA	NA	NA	NA
	idsat(uA/um)	-5.41E+01	-1.29E+02	-1.27E+02	-1.28E+02	-5.60E+01	-1.27E+02	-1.18E+02	-1.15E+02
	vtlin(V)	-4.50E-01	-4.46E-01	-4.44E-01	-4.42E-01	-4.44E-01	-4.44E-01	-4.50E-01	-4.52E-01
	vtsat(V)	-4.12E-01	-3.81E-01	-3.82E-01	-3.81E-01	-4.09E-01	-3.77E-01	-3.85E-01	-3.89E-01
	body_effect(V)	-3.03E-01	-2.52E-01	-2.50E-01	-2.49E-01	-3.02E-01	-2.52E-01	-2.51E-01	-2.50E-01
	ioff(pA/um)	-4.42E+01	-3.17E+02	-2.78E+02	-2.61E+02	-4.67E+01	-3.43E+02	-2.58E+02	-2.23E+02
FNFP	ft(GHz)	10.275	34.581	28.411	27.61	NA	NA	NA	NA
	idsat(uA/um)	-5.13E+01	-1.16E+02	-1.13E+02	-1.14E+02	-5.45E+01	-1.16E+02	-1.07E+02	-1.04E+02
	vtlin(V)	-4.77E-01	-4.79E-01	-4.81E-01	-4.81E-01	-4.71E-01	-4.77E-01	-4.86E-01	-4.91E-01
	vtsat(V)	-4.38E-01	-4.15E-01	-4.19E-01	-4.20E-01	-4.36E-01	-4.11E-01	-4.22E-01	-4.28E-01
	body_effect(V)	-3.03E-01	-2.54E-01	-2.52E-01	-2.51E-01	-3.02E-01	-2.54E-01	-2.52E-01	-2.51E-01
	ioff(pA/um)	-2.42E+01	-1.50E+02	-1.22E+02	-1.09E+02	-2.57E+01	-1.65E+02	-1.15E+02	-9.47E+01
FF	ft(GHz)	10.417	38.245	33.291	32.744	NA	NA	NA	NA
	idsat(uA/um)	-6.22E+01	-1.55E+02	-1.55E+02	-1.58E+02	-6.20E+01	-1.50E+02	-1.41E+02	-1.39E+02
	vtlin(V)	-3.98E-01	-3.85E-01	-3.76E-01	-3.70E-01	-3.93E-01	-3.83E-01	-3.82E-01	-3.79E-01
	vtsat(V)	-3.62E-01	-3.21E-01	-3.15E-01	-3.10E-01	-3.60E-01	-3.18E-01	-3.19E-01	-3.18E-01
	body_effect(V)	-2.95E-01	-2.44E-01	-2.43E-01	-2.42E-01	-2.94E-01	-2.45E-01	-2.43E-01	-2.43E-01
	ioff(pA/um)	-1.31E+02	-1.03E+03	-1.08E+03	-1.13E+03	-1.37E+02	-1.09E+03	-9.83E+02	-9.50E+02

Table 2-6 RF & Logic Corner Table at 125 °C

Temp=-55C		RF Model				Logic Model			
Device	LF (um)	0.24	0.12	0.12	0.12	0.24	0.12	0.12	0.12
	WF (um)	9.6	9.6	2.4	1.6	9.6	9.6	2.4	1.6
	NF	22	16	4	16	22	16	4	16
SS	ft(GHz)	11.118	38.807	31.084	30.031	NA	NA	NA	NA
	idsat(uA/um)	-4.46E+01	-1.01E+02	-9.81E+01	-9.80E+01	-4.84E+01	-1.03E+02	-9.39E+01	-9.03E+01
	vtlin(V)	-6.47E-01	-6.61E-01	-6.65E-01	-6.67E-01	-6.44E-01	-6.60E-01	-6.70E-01	-6.76E-01
	vtsat(V)	-6.21E-01	-6.06E-01	-6.12E-01	-6.15E-01	-6.18E-01	-6.01E-01	-6.14E-01	-6.21E-01
	body_effect(V)	-2.85E-01	-2.37E-01	-2.35E-01	-2.34E-01	-2.84E-01	-2.37E-01	-2.35E-01	-2.34E-01
	ioff(pA/um)	-4.89E-02	-4.36E-02	-1.61E-01	-2.04E-01	-2.76E-02	-2.80E-02	-1.11E-01	-1.67E-01
SNFP	ft(GHz)	11.993	45.266	38.779	38.109	NA	NA	NA	NA
	idsat(uA/um)	-5.92E+01	-1.50E+02	-1.50E+02	-1.53E+02	-6.16E+01	-1.49E+02	-1.41E+02	-1.38E+02
	vtlin(V)	-5.70E-01	-5.69E-01	-5.63E-01	-5.58E-01	-5.67E-01	-5.68E-01	-5.68E-01	-5.67E-01
	vtsat(V)	-5.45E-01	-5.12E-01	-5.09E-01	-5.05E-01	-5.43E-01	-5.09E-01	-5.12E-01	-5.13E-01
	body_effect(V)	-2.76E-01	-2.24E-01	-2.23E-01	-2.21E-01	-2.75E-01	-2.25E-01	-2.23E-01	-2.22E-01
	ioff(pA/um)	-9.73E-02	-8.66E-02	-2.13E-01	-2.66E-01	-2.82E-02	-3.93E-02	-1.19E-01	-1.74E-01
TT	ft(GHz)	11.688	42.719	35.958	35.188	NA	NA	NA	NA
	idsat(uA/um)	-5.46E+01	-1.34E+02	-1.33E+02	-1.35E+02	-5.66E+01	-1.33E+02	-1.24E+02	-1.21E+02
	vtlin(V)	-5.96E-01	-6.00E-01	-5.97E-01	-5.94E-01	-5.93E-01	-5.98E-01	-6.02E-01	-6.03E-01
	vtsat(V)	-5.70E-01	-5.43E-01	-5.42E-01	-5.41E-01	-5.68E-01	-5.39E-01	-5.46E-01	-5.49E-01
	body_effect(V)	-2.76E-01	-2.25E-01	-2.23E-01	-2.22E-01	-2.75E-01	-2.25E-01	-2.24E-01	-2.23E-01
	ioff(pA/um)	-9.71E-02	-7.97E-02	-2.07E-01	-2.59E-01	-2.78E-02	-3.17E-02	-1.13E-01	-1.68E-01
FNFP	ft(GHz)	11.322	39.756	32.831	31.958	NA	NA	NA	NA
	idsat(uA/um)	-5.00E+01	-1.17E+02	-1.15E+02	-1.16E+02	-5.27E+01	-1.17E+02	-1.08E+02	-1.05E+02
	vtlin(V)	-6.22E-01	-6.32E-01	-6.32E-01	-6.32E-01	-6.19E-01	-6.30E-01	-6.37E-01	-6.41E-01
	vtsat(V)	-5.96E-01	-5.76E-01	-5.79E-01	-5.80E-01	-5.94E-01	-5.72E-01	-5.82E-01	-5.86E-01
	body_effect(V)	-2.76E-01	-2.27E-01	-2.25E-01	-2.24E-01	-2.75E-01	-2.28E-01	-2.26E-01	-2.25E-01
	ioff(pA/um)	-9.71E-02	-7.75E-02	-2.04E-01	-2.57E-01	-2.76E-02	-2.88E-02	-1.11E-01	-1.67E-01
FF	ft(GHz)	12.073	45.05	39.648	39.206	NA	NA	NA	NA
	idsat(uA/um)	-6.53E+01	-1.67E+02	-1.69E+02	-1.74E+02	-6.57E+01	-1.63E+02	-1.56E+02	-1.54E+02
	vtlin(V)	-5.46E-01	-5.41E-01	-5.31E-01	-5.24E-01	-5.43E-01	-5.40E-01	-5.36E-01	-5.33E-01
	vtsat(V)	-5.21E-01	-4.85E-01	-4.77E-01	-4.72E-01	-5.20E-01	-4.82E-01	-4.82E-01	-4.80E-01
	body_effect(V)	-2.68E-01	-2.17E-01	-2.15E-01	-2.14E-01	-2.66E-01	-2.17E-01	-2.15E-01	-2.14E-01
	ioff(pA/um)	-2.69E-01	-2.21E-01	-3.86E-01	-4.70E-01	-2.84E-02	-5.47E-02	-1.34E-01	-1.88E-01

Table 2-7 RF & Logic Corner Table at -55 °C

2.2.2 Summary Table

In the following fitting plots & summary table, the parasitic resistances due to metal interconnection from pad to DUT (Device Under Test) are taken into account. Since RF DUT generally use larger total gate width and has larger current, these parasitic resistances will cause voltage drops. Therefore the extracted I_{dsat} is smaller than the real I_{dsat} for the DUT without the parasitic resistance.

Device						Vtlin(V)		Vtsat(V)		Idsat(uA/um)		Ioff(pA/um)		ft(GHz)	
Name	LF(um)	WF(um)	NF	M	con	Measure	Model	Measure	Model	Measure	Model	Measure	Model	Measure	Model
PFET601	0.12	1.6	16	6	1	-0.531	-0.519	-0.467	-0.463	-125.49	-126.94	-3.52	-2.474	30.235	30.955
PFET602	0.12	2.4	16	4	1	-0.531	-0.521	-0.466	-0.464	-127.05	-125.57	-3.56	-2.596	33.082	33.198
PFET603	0.12	4.8	16	2	1	-0.534	-0.523	-0.469	-0.465	-127.49	-124.83	-3.1	-2.79	36.895	35.674
PFET604	0.12	9.6	16	1	1	-0.533	-0.523	-0.468	-0.464	-124.86	-124.48	-3.62	-2.948	37.876	36.955
PFET605	0.15	2.4	16	4	1	-0.539	-0.525	-0.499	-0.482	-90.16	-89.002	-1.15	-0.837	21.382	21.764
PFET606	0.18	2.4	16	4	1	-0.534	-0.524	-0.502	-0.488	-72.43	-71.502	-0.78	-0.569	15.282	15.79
PFET607	0.24	2.4	16	4	1	-0.528	-0.521	-0.5	-0.491	-53.26	-52.835	-0.82	-0.441	9.268	9.52
PFET608	0.36	2.4	16	4	1	-0.521	-0.516	-0.496	-0.489	-35.01	-36.085	-0.58	-0.385	4.514	4.676
PFET609	0.12	2.4	2	18	1	-0.534	-0.521	-0.475	-0.464	-129.51	-126.23	-2.55	-2.645	30.44	30.154
PFET611	0.12	2.4	5	13	1	-0.533	-0.521	-0.47	-0.464	-127.83	-126.28	-3.03	-2.612	32.429	32.307
PFET612	0.12	2.4	8	8	1	-0.533	-0.521	-0.47	-0.464	-127.24	-125.73	-3.45	-2.603	32.493	32.727
PFET613	0.12	2.4	11	6	1	-0.532	-0.521	-0.467	-0.464	-127.03	-125.75	-3.6	-2.599	32.842	33.024
PFET619	0.12	4.8	4	8	1	-0.534	-0.523	-0.47	-0.465	-130.31	-125.35	-2.96	-2.8	36.622	34.877
PFET620	0.12	4.8	11	2	1	-0.531	-0.523	-0.466	-0.465	-128.98	-124.86	-3.94	-2.791	36.94	35.63
PFET621	0.12	4.8	22	1	1	-0.533	-0.523	-0.468	-0.465	-127.55	-125.32	-3.79	-2.788	37.154	35.92
PFET622	0.15	9.6	4	4	1	-0.539	-0.527	-0.501	-0.483	-94.69	-89.105	-1.14	-0.798	25.293	24.177
PFET623	0.18	9.6	11	1	1	-0.536	-0.527	-0.505	-0.49	-74.19	-71.43	-1.07	-0.484	17.911	17.681
PFET624	0.24	9.6	22	1	1	-0.533	-0.524	-0.505	-0.493	-53.29	-52.56	-0.88	-0.335	10.537	10.578
PFET625	0.12	2.4	4	1	1	-0.529	-0.521	-0.468	-0.464	-133.63	-128.28	-7.11	-2.617	35.494	32.591

Table 2-8 Summary Table for RF PMOS

Measure: Measurement result

Model: Simulation result

2.2.3 Simulation results for WF, LF trends

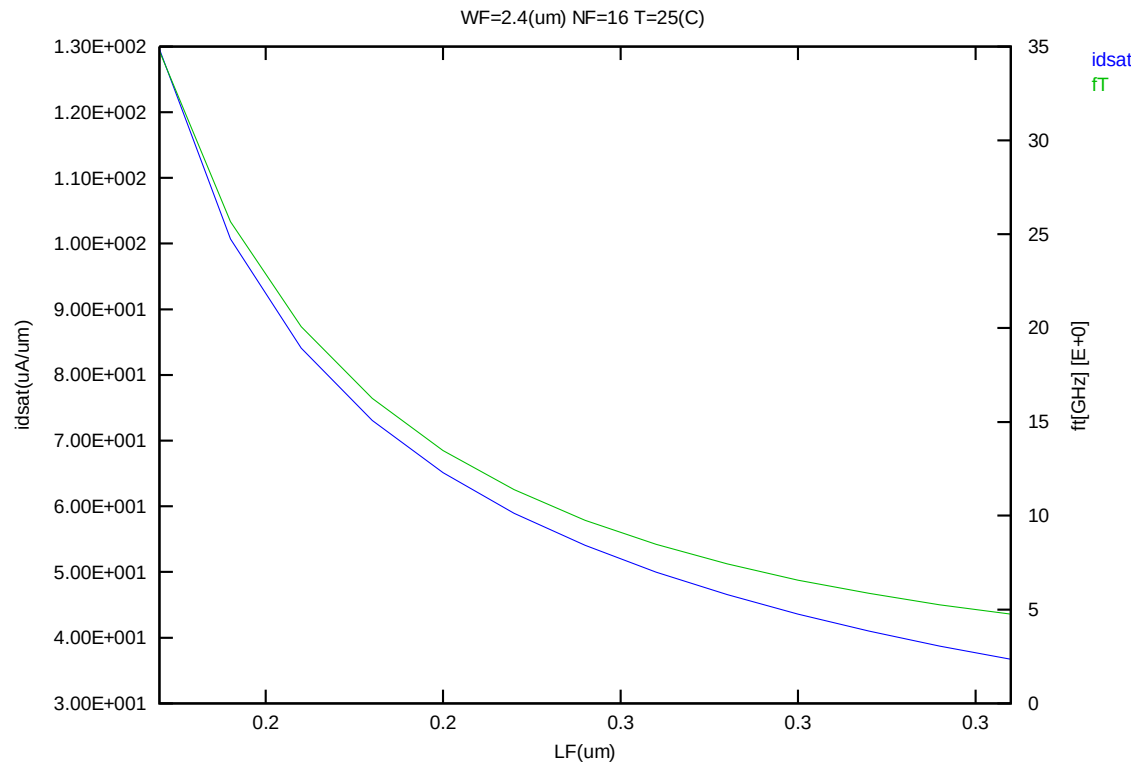


Figure 2-14 fT and Idsat vs. channel length

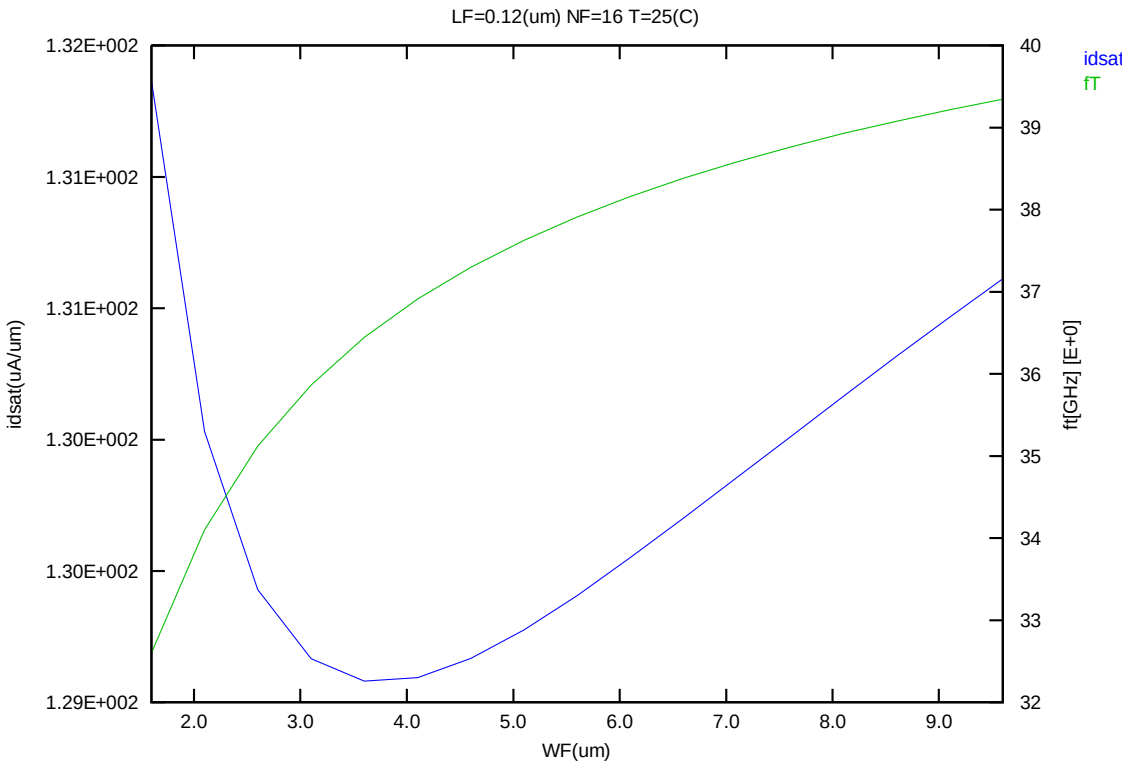


Figure 2-15 fT and Idsat vs. channel width

2.2.4 DC & RF Fitting curves for individual devices

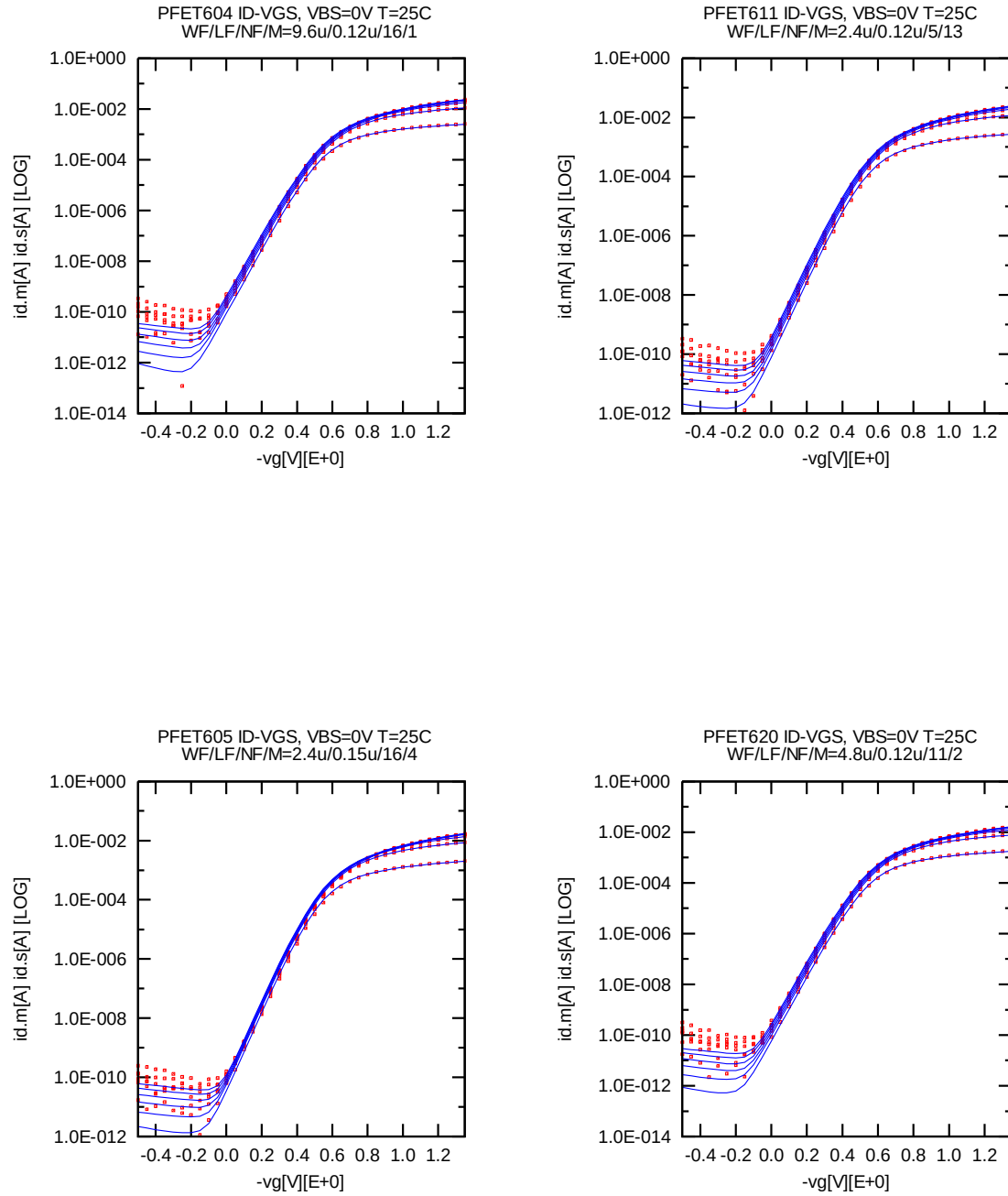


Figure 2-16 Id-Vg (25°C)

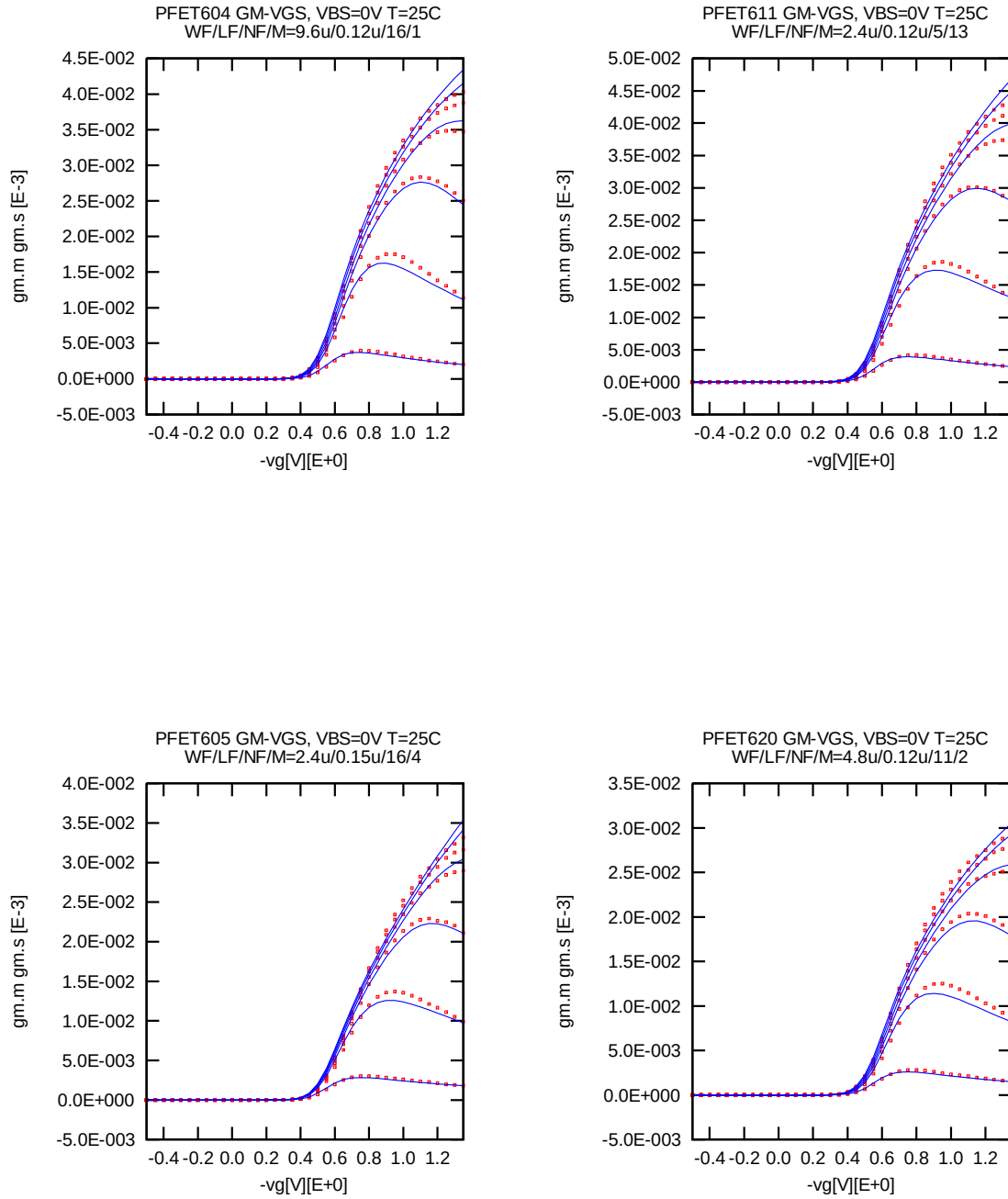


Figure 2-17 Gm-Vg (25°C)

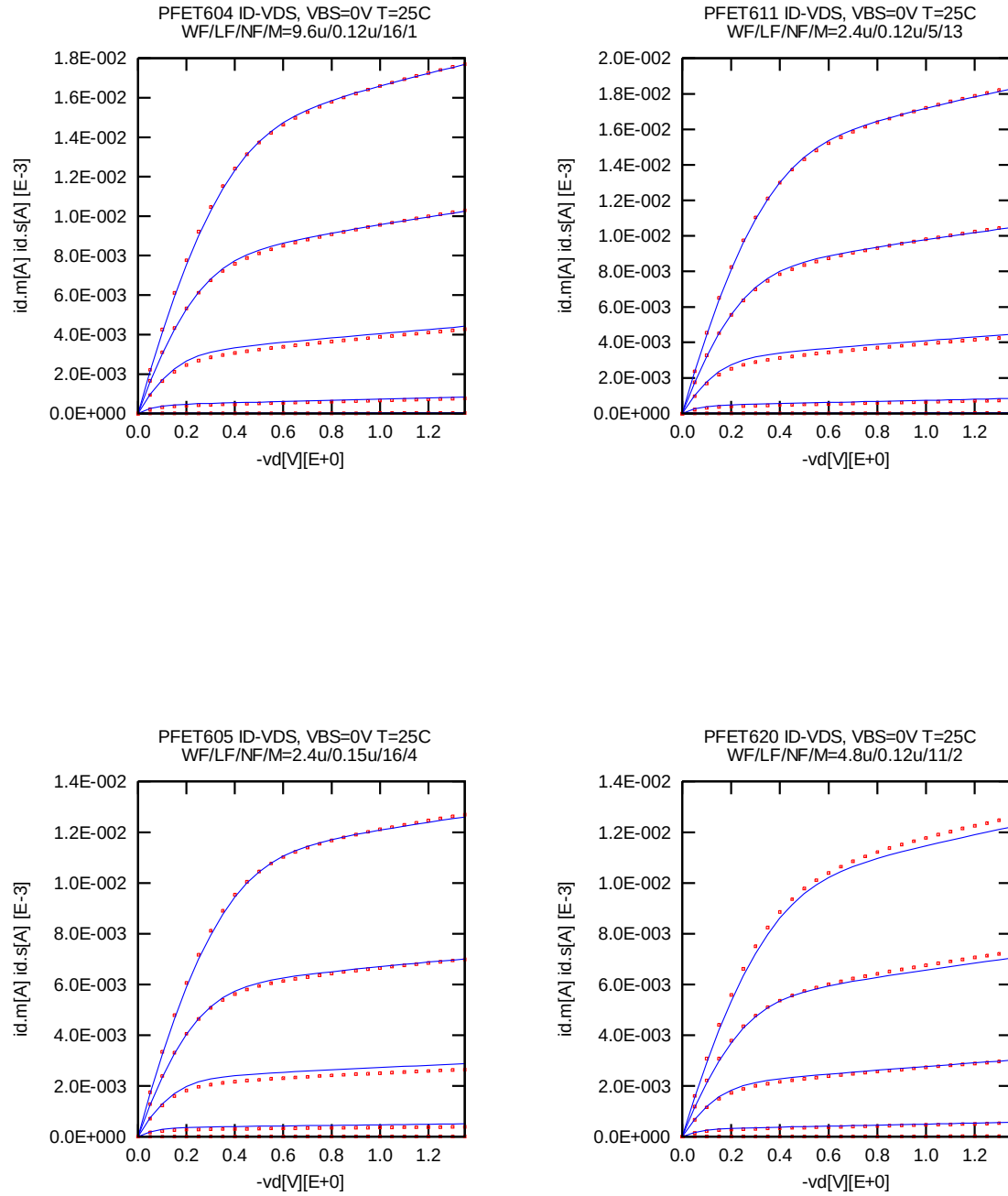


Figure 2-18 Id-Vd (25°C)

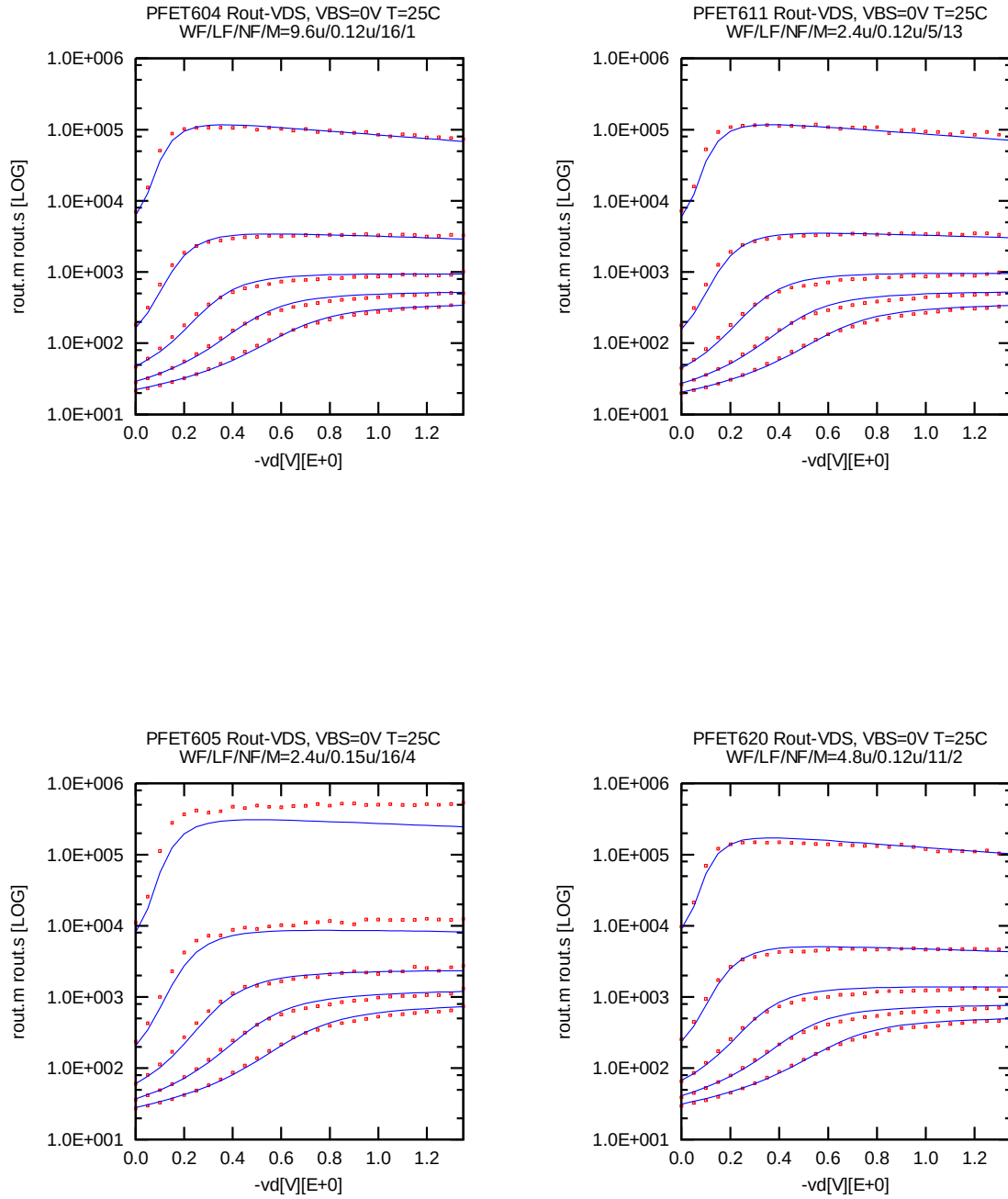


Figure 2-19 Rout-Vd (25°C)

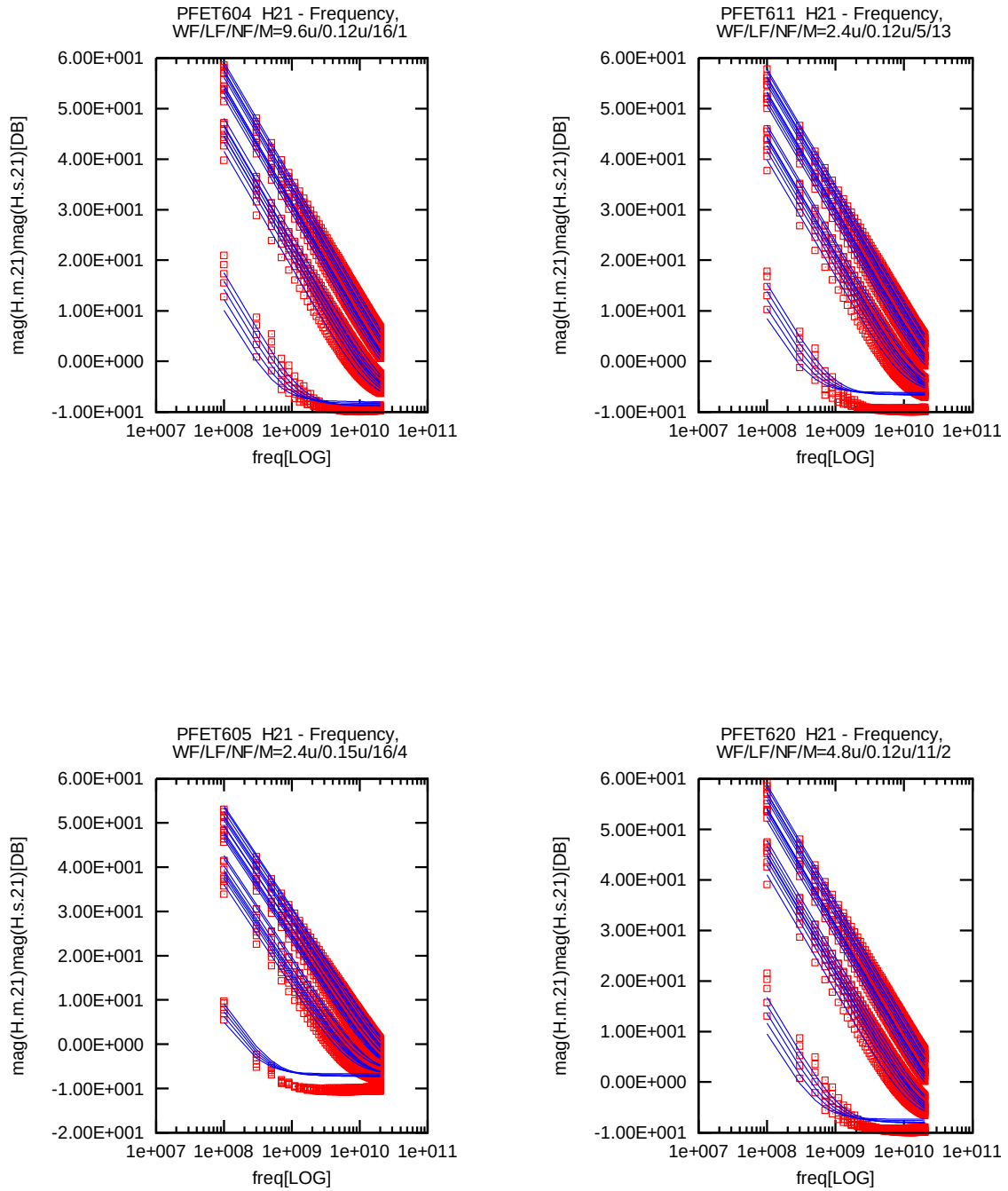


Figure 2-20 Mag(H21)-Freq (25°C)

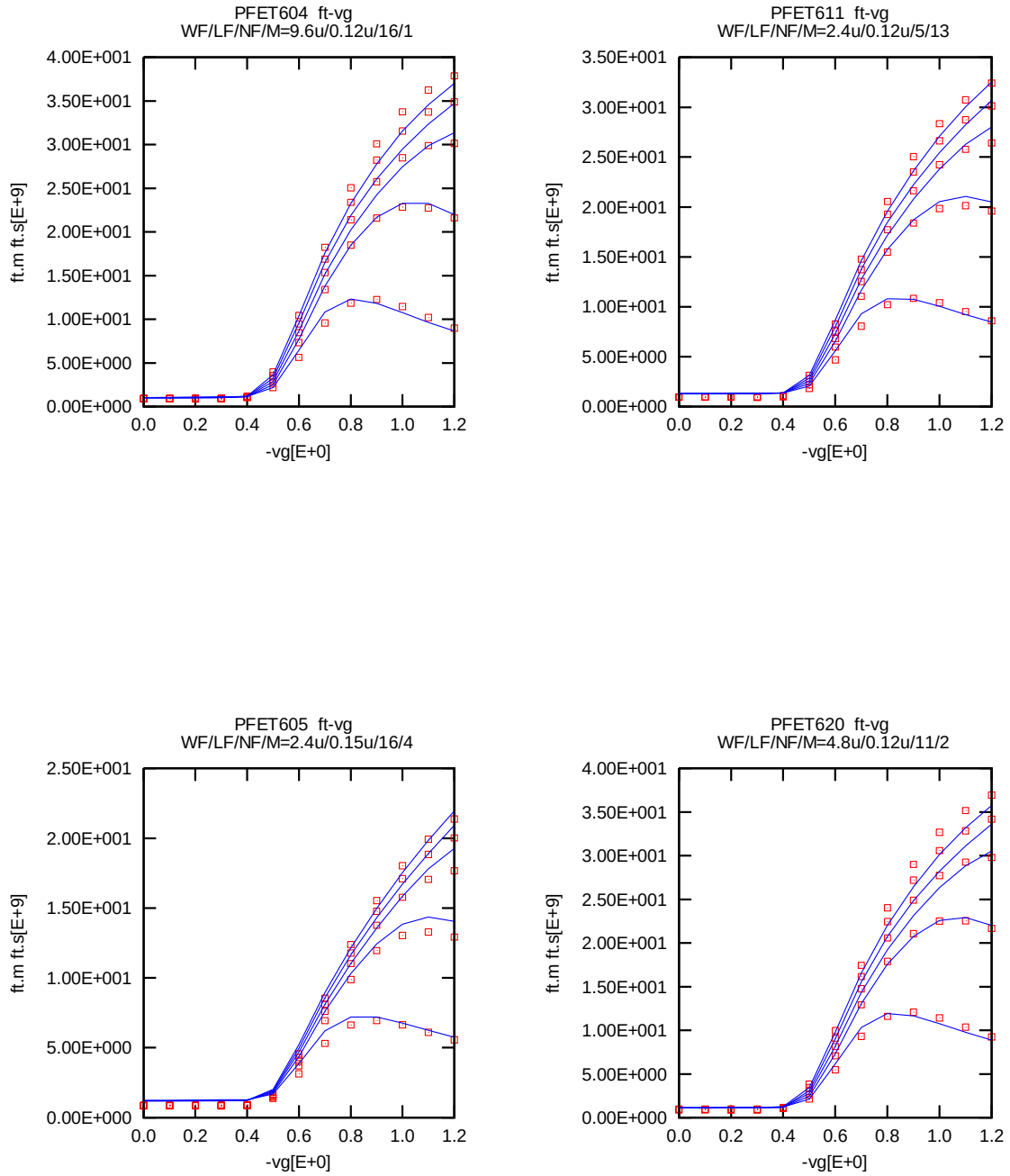


Figure 2-21 f_T - V_g (25°C)

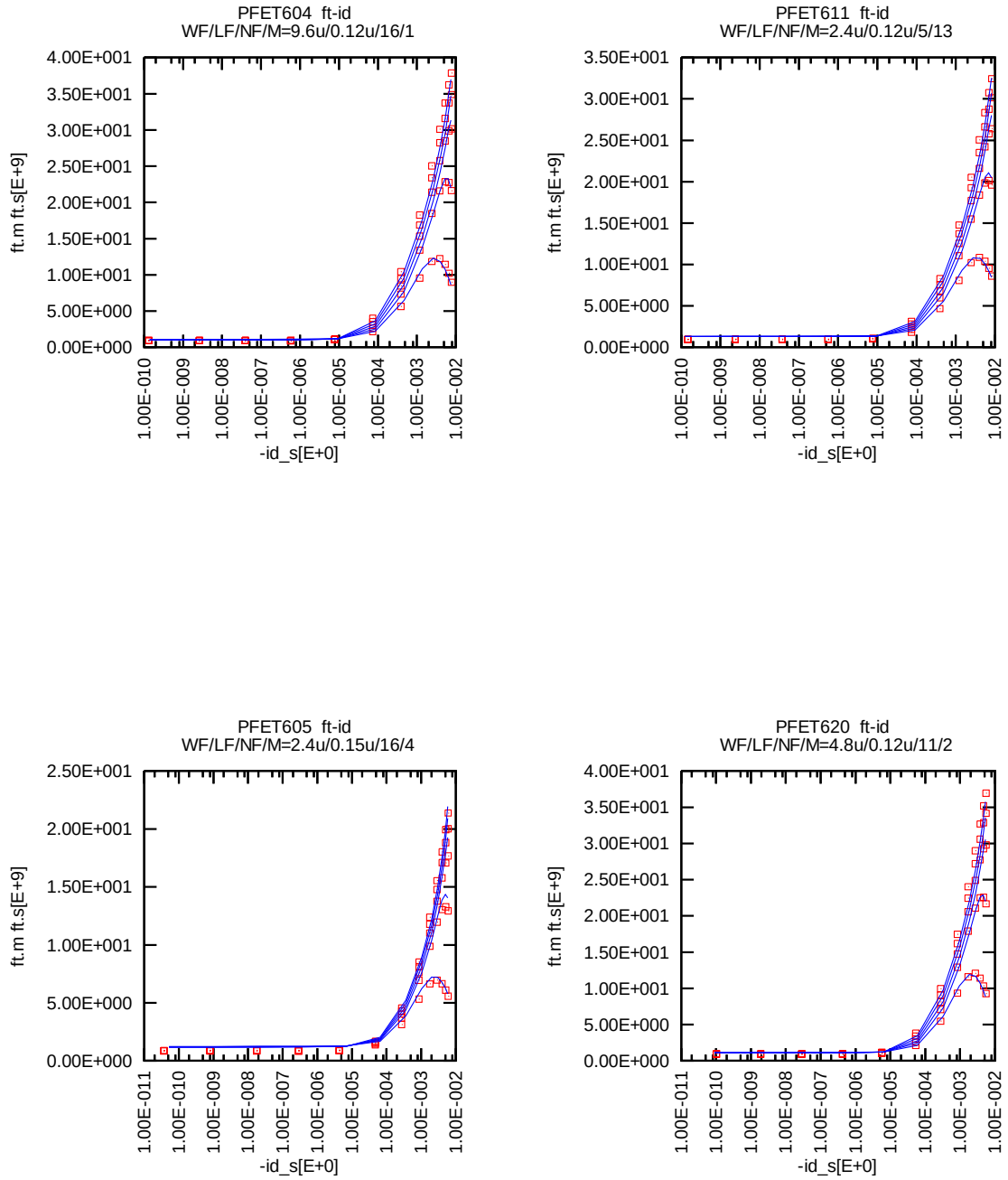
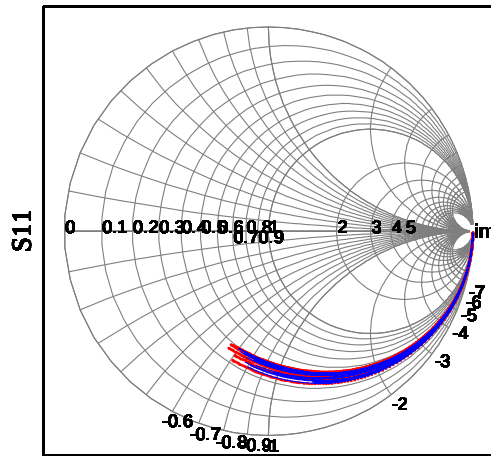
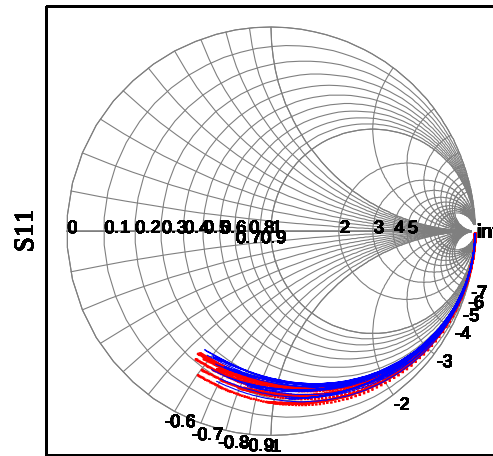


Figure 2-22 f_{T-id} (25°C)

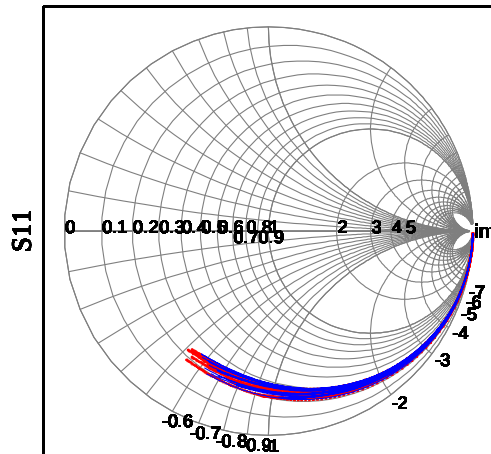
PFET604 S11 WF/LF/NF/M=9.6u/0.12u/16/1



PFET611 S11 WF/LF/NF/M=2.4u/0.12u/5/13



PFET605 S11 WF/LF/NF/M=2.4u/0.15u/16/4



PFET620 S11 WF/LF/NF/M=4.8u/0.12u/11/2

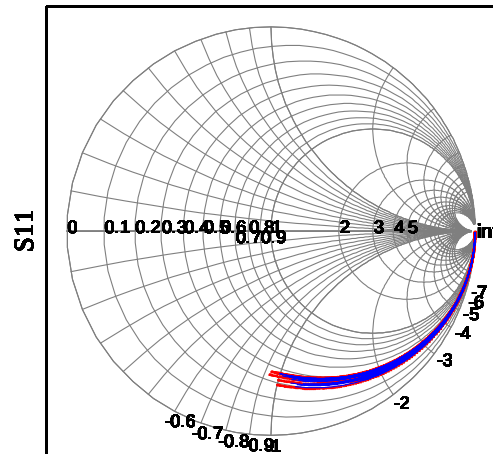


Figure 2-23 S11 (25°C)

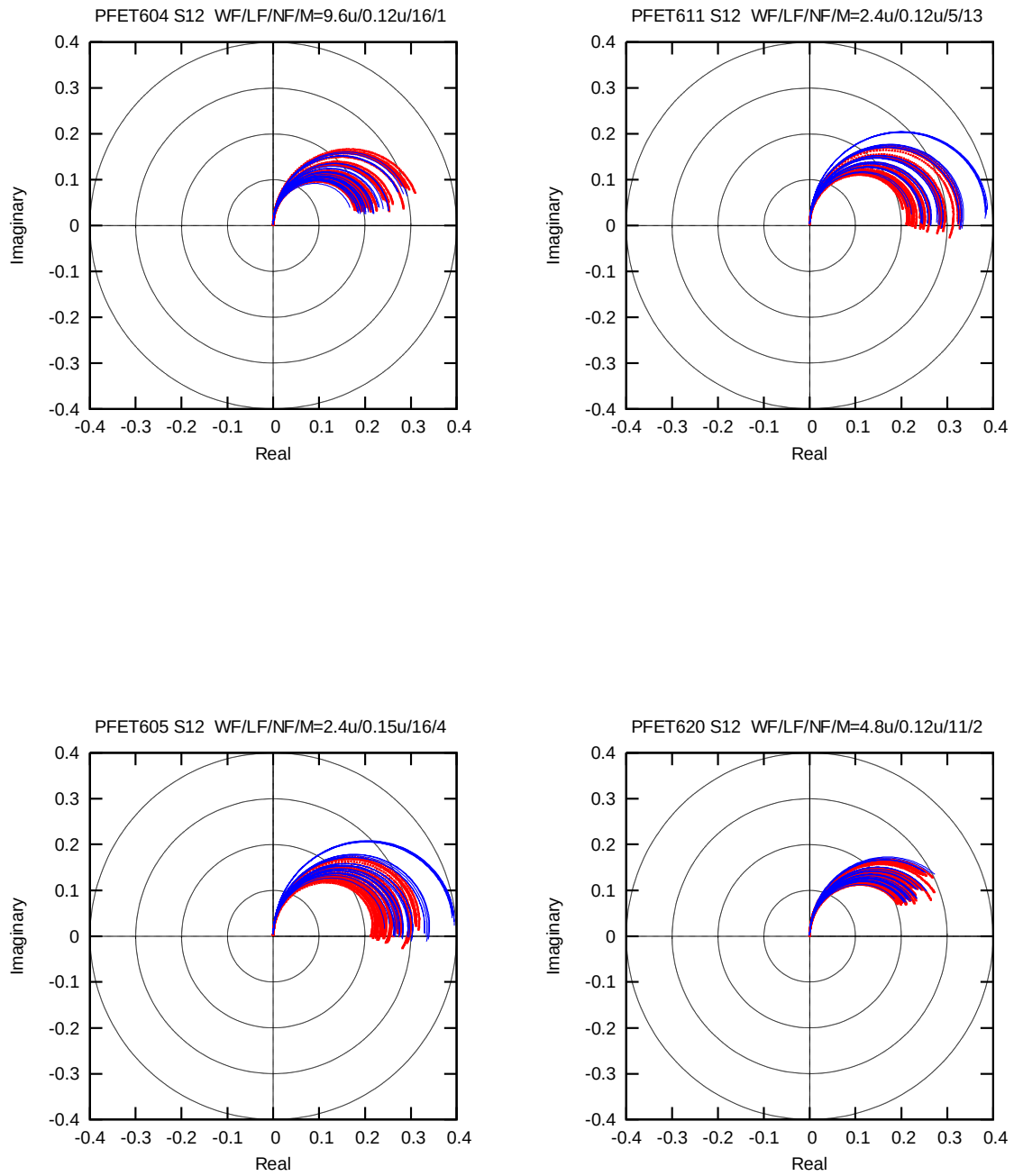


Figure 2-24 S12 (25°C)

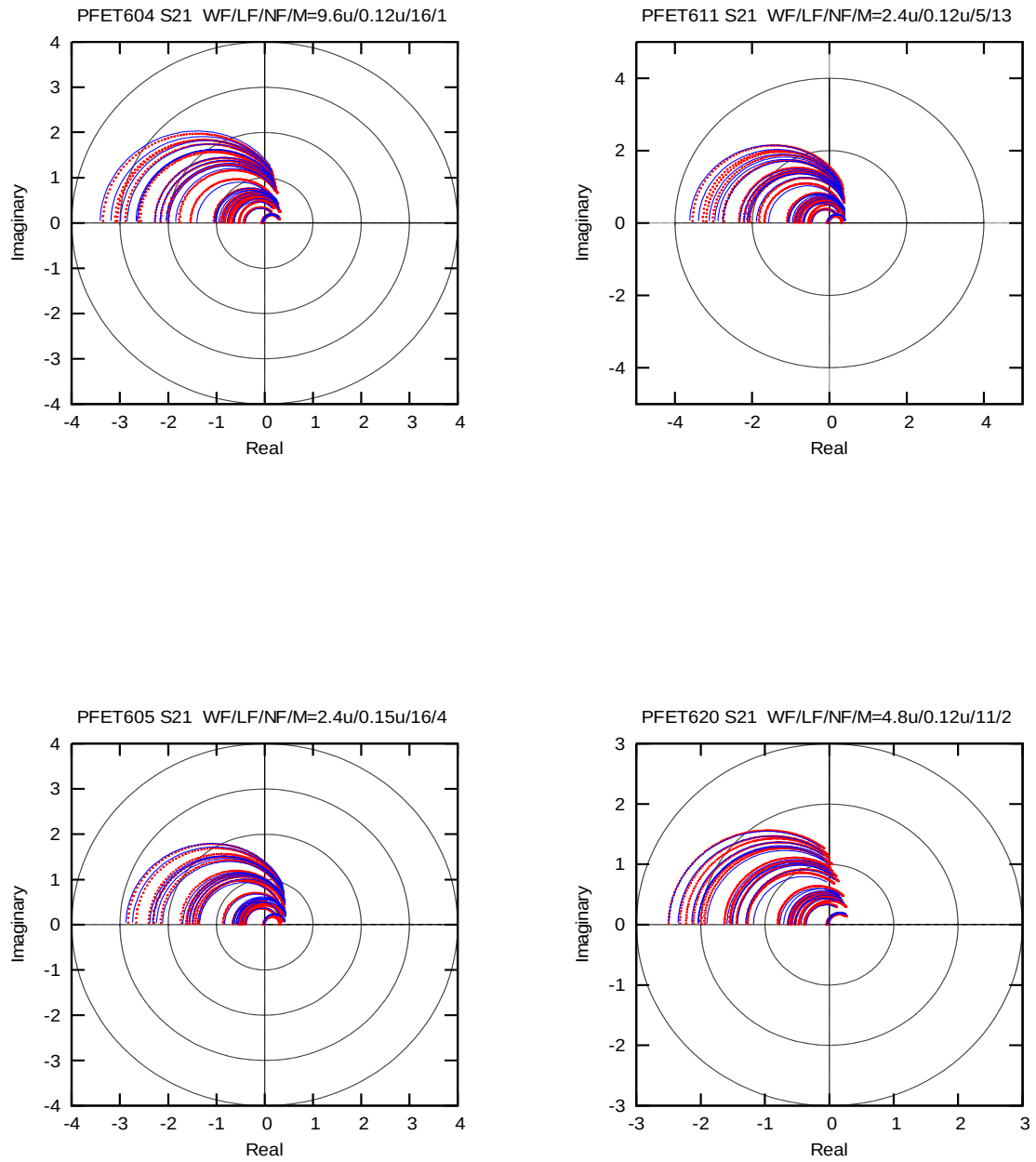
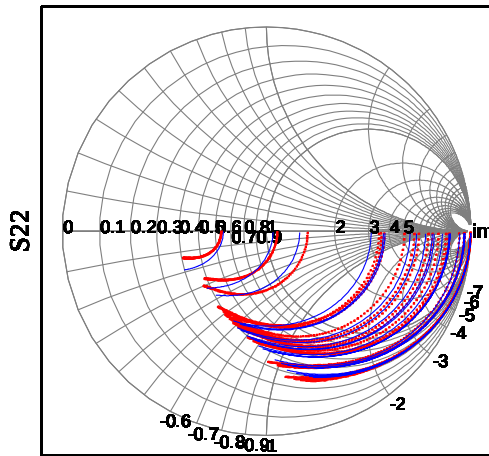
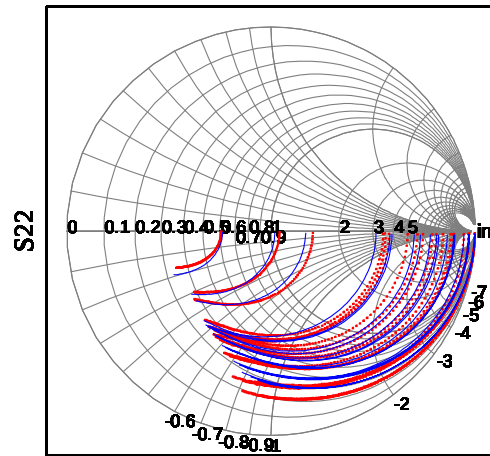


Figure 2-25 S21 (25°C)

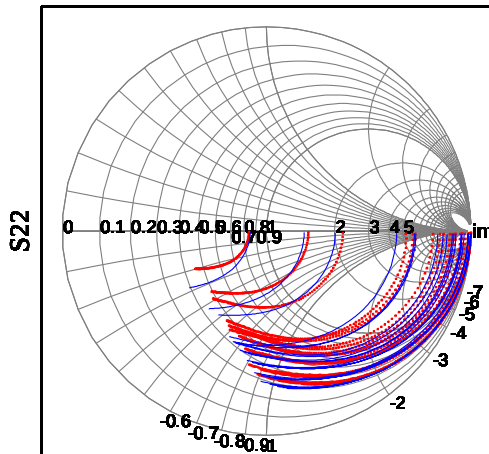
PFET604 S22 WF/LF/NF/M=9.6u/0.12u/16/1



PFET611 S22 WF/LF/NF/M=2.4u/0.12u/5/13



PFET605 S22 WF/LF/NF/M=2.4u/0.15u/16/4



PFET620 S22 WF/LF/NF/M=4.8u/0.12u/11/2

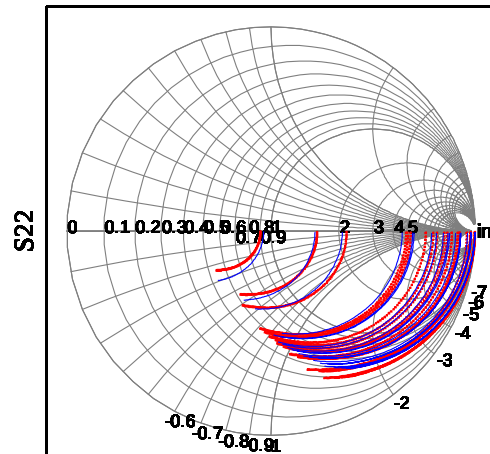


Figure 2-26 S22 (25°C)

3 Appendix

3.1 HSPICE Warning Message

Table 3-1 HSPICE warning message

NO.	Warning Message	Explanation
1.	None	None

3.2 SPECTRE Warning Message

Table 3-2 SPECTRE warning message

NO.	Warning Message	Explanation
1.	None	None

3.3 ELDO Warning Message

Table 3-3 ELDO warning message

NO.	Warning Message	Explanation
1.	None	None